



**“Interpretable Deep Multimodal Prediction of 30-Day
Readmissions in Elderly Patients Using Fine-Tuned
ClinicalBERT and EHR Data”**

Mark Mckeen C21440394

**Submitted in partial fulfilment of the requirements for the
degree of:**

**BSc in Business Analytics
Technological University Dublin**

Supervisor: Thoa Pham

Submission Date: 01/05/2025

Word Count: 15,986

Declaration

This dissertation is submitted by the undersigned to the Technological University Dublin in partial fulfilment for the degree Business Analytics. It is entirely the author's own work and has not been submitted previously for an award to this or any other institution. All sources consulted are appropriately referenced as per the TUD School of Business Style Guide.

Signed: Mark Mckeon

Date: 01/05/2025

Acknowledgements

I would like to express my sincere gratitude to my academic supervisor, Thoa Pham, for sharing her knowledge, insights, and continued assistance throughout this academic year. To my classmates, for their encouragement and support throughout the last four years. Finally, to my parents, for their continuous support and for all the opportunities they have provided me with.

Abstract

Hospital readmissions among elderly patients remain a persistent challenge, driving healthcare costs and straining hospital capacity. This study aims to predict 30-day readmissions in older adults using advanced machine learning and deep learning models applied to structured and unstructured data from the MIMIC-IV electronic health record (EHR) database. Baseline models, including Logistic Regression, Random Forest, and XGBoost, were developed using structured EHR features. A structured-data Multi-Layer Perceptron (MLP) was also trained, achieving an AUROC of 0.6656. ClinicalBERT was fine-tuned on section-filtered discharge summaries to incorporate narrative clinical insights using a sliding window and attention pooling approach. A lightweight MLP classifier trained on the resulting [CLS] embeddings achieved an AUROC of 0.7418. Two fusion strategies were implemented: a mid-fusion model combining structured features and ClinicalBERT embeddings via gated attention (AUROC: 0.7517), and a late-fusion Random Forest meta-classifier ensembling probability outputs from the MLP models, which achieved the best performance (AUROC: 0.7608, F1: 0.4774). Key enhancements such as SMOTEENN resampling, focal loss, and threshold optimisation improved model sensitivity to high-risk cases. These findings demonstrate that multimodal AI models can significantly improve early identification of readmission risk while maintaining clinical interpretability. By enabling proactive, cost-effective intervention, this research supports the United Nations Sustainable Development Goals SDG 3 (Good Health and Well-Being) through improved care continuity and SDG 13 (Climate Action) by reducing avoidable hospital utilization and associated environmental impacts.

Table of Contents

Chapter 1 - Introduction	11
1.1 Introduction.....	11
1.2 Research Question.....	11
1.4 Analysis Framework.....	12
1.5 Structure of Dissertation.....	13
Chapter 2 - Literature Review	13
2.1 Hospital Readmission Prediction	13
2.1.1 Key Clinical and Demographic Factors for 30-Day Readmissions.....	14
2.2 Preprocessing in Predictive Modelling	15
2.2.1 Handling Missing Data.....	15
2.2.2 Feature Engineering and Selection	15
2.2.3 Addressing Data Imbalance.....	16
2.3 Machine Learning (ML) Models in Predictive Analytics.....	16
2.3.1 Traditional Predictive Methods	16
2.3.2 Modern Machine Learning Techniques	17
2.3.3 Tree-Based Models	17
2.3.4 Ensemble Methods.....	17
2.3.5 Deep Learning Models	18
2.3.6 Multi-Modal and Hybrid Models	18
2.3.7 Model Performance Across Studies	19
2.4 Leveraging Electronic Health Record (EHR) Data	19
2.4.1 Structured vs. Unstructured Data	19
2.4.2 Key Predictors from EHR Data	20
2.4.3 Natural Language Processing (NLP) for Unstructured Data	20
2.4.4 Challenges of EHR Data.....	21
2.5 Model Evaluation	21
2.5.1 Key Metrics for Evaluation	22
2.5.2 Model Interpretability.....	23

2.5.3 Validation Techniques	24
2.5.4 Model Performance	25
2.5.5 Challenges in Evaluation	26
2.6 Challenges and Limitations	26
2.6.1 Data Challenges.....	26
2.6.2 Model Challenges	27
2.6.3 Operational Challenges	27
2.7 Research Gaps	28
2.7.1 Enhancing Data Quality and Accessibility	28
2.7.2 Advancing Model Development	29
2.7.3 Expanding Data Modalities	29
2.7.4 Addressing Operational and Ethical Challenges.....	30
2.7.5 Rigorous Validation and Benchmarking	30
2.8 Conclusion	30
Chapter 3 - Methodology.....	32
3.1 Data Acquisition.....	32
3.1.1 Structured Dataset Overview	33
3.2 Data Preprocessing	34
3.2.1 Handling of Large-Scale Datasets NB.....	36
3.2.2 Data Transformation and Data Filtering	37
3.2.3 Data Merging	37
3.2.4 Data Cleaning and Imputation.....	37
3.2.5 Feature Engineering	38
3.2.6 Encoding of Categorical and Code-Based Variables	39
3.2.7 Final Scaling and Alignment.....	40
3.2.8 Text Data Processing.....	40
3.2.9 Class Balancing	40
3.2.10 Cross-Cutting Preprocessing Strategies	41
3.3 Exploratory Data Analysis (EDA).....	41

3.3.1 Patient Demographics and Cohort Overview	41
3.3.2 Feature Distributions and Correlations	43
3.3.3 Categorical Predictors and Readmission Risk	44
3.3.4 Diagnoses and Procedures	46
3.3.5 Clinical Readmission Patterns	47
3.4 Modelling.....	49
3.4.1 Baseline Models on Structured Data	50
3.4.1.1 Data Preparation	51
3.4.1.2 Logistic Regression (LR)	51
3.4.1.3 Random Forest (RF)	52
3.4.1.4 XGBoost (XGB).....	52
3.4.1.5 Evaluation	53
3.4.2 Deep Learning.....	53
3.4.2.1 MLP on Structured Data	53
3.4.2.2 Fine-Tuned ClinicalBERT on 30-day Readmission	55
3.4.2.3 MLP Classifier on ClinicalBERT Embeddings	58
3.4.2.4 Mid-Fusion Models	59
3.4.2.5 Late Fusion Model	61
3.5 XAI and Feature Importance Approaches.....	62
Chapter 4 - Findings and Results	64
4.1 Evaluation Metrics	64
4.2 Baseline Model Results (Structured Data Only)	65
4.3 Deep Learning Models Results	68
4.4 Fusion Models Results	71
4.5 Explainability and Feature Importance Results	73
4.6 Final Comparison and Analysis	82
4.7 Clinical Relevance	83
4.8 Conclusion of Findings and Results.....	84
Chapter 5 - Discussion.....	85

5.1 Summary of Key Findings.....	85
5.2 Discussion of Key Results	85
5.3 Implications for Clinical Practice.....	87
5.4 Limitations of the Study	88
5.5 Future Work	89
5.6 Summary of Discussion	90
Chapter 6 - Conclusion	91
6.1 Restatement of Research Purpose	91
6.2 Major Contributions.....	91
6.3 Broader Implications	91
6.4 Final Reflections.....	93
Bibliography	94
Appendix	100

List of Figures

Figure 1. Analysis Framework.....	13
Figure 2. Overview of the data preprocessing pipeline for structured EHR features and unstructured discharge summaries.....	36
Figure 3. Age distribution of elderly patient admissions (aged 65+) in the study cohort.	42
Figure 4. Proportion of admissions resulting in 30-day readmission versus no readmission.	43
Figure 5. Correlation heatmap of selected numerical features in the structured dataset.....	44
Figure 6. 30-day readmission rate by admission type across the study cohort.	45
Figure 7. 30-day readmission rate by admission location at time of hospitalization.	45
Figure 8. 30-day readmission rate by top 10 most common individual ICD-9 diagnoses.	46
Figure 9. 30-day readmission rate by top 10 most common individual ICD-9 procedures.....	47
Figure 10. 30-day readmission rate stratified by presence of polypharmacy (≥ 5 concurrent medications).	48
Figure 11. Average unique drug count among readmitted versus non-readmitted patients.	48
Figure 12. Proportion of abnormal lab results stratified by 30-day readmission status.....	49
Figure 13. Baseline classical machine learning modelling pipeline for Logistic Regression, Random Forest, and XGBoost models.	50
Figure 14. Structured MLP modelling pipeline: feature selection, SMOTEENN resampling, standardization, MLP setup, and evaluation.	54

Figure 15. Illustration of a basic Multi-Layer Perceptron (MLP) architecture with input, hidden, and output layers.	55
Figure 16. Fine-tuning pipeline for ClinicalBERT on section-filtered discharge summaries for 30-day readmission prediction.	57
Figure 17. ClinicalBERT architecture for extracting biomedical entity relationships from clinical notes (Khetan et al., 2021).	58
Figure 18. Pipeline for MLP classifier training on mean-pooled [CLS] embeddings from fine-tuned ClinicalBERT.....	59
Figure 19. Attention-based mid-fusion modelling pipeline combining structured features and ClinicalBERT embeddings with dynamic modality gating.	60
Figure 20. Late fusion modelling pipeline using a Random Forest meta-classifier to combine model outputs.	62
Figure 21. Formula for computing the F1 score, the harmonic mean of precision and recall.	65
Figure 22. Illustration of perfect and random ROC curves, demonstrating AUC = 1.0 and AUC = 0.5 performance scenarios.....	65
Figure 23. ROC Curve comparing baseline models (Logistic Regression, Random Forest, XGBoost) on structured data.	67
Figure 24. Precision-Recall Curve comparing baseline models (Logistic Regression, Random Forest, XGBoost) on structured data.	68
Figure 25. ROC Curve for fine-tuned ClinicalBERT model on predicting 30-day readmissions.....	69
Figure 26. Precision-Recall Curve for fine-tuned ClinicalBERT model on predicting 30-day readmissions.	70
Figure 27. ROC Curve comparing all models (Structured MLP, ClinicalBERT MLP, Late Fusion RF, Hybrid Fusion MLP).	72
Figure 28. Precision-Recall Curve comparing all models (Structured MLP, ClinicalBERT MLP, Late Fusion RF, Hybrid Fusion MLP).....	73
Figure 29. SHAP Feature Importance for Structured MLP.	75
Figure 30. Integrated Gradients - Top Embedding Dimensions (ClinicalBERT MLP).	76
Figure 31. Saliency Visualization of Notes - Example 1.	77
Figure 32. Saliency Visualization of Notes - Example 2.	78
Figure 33. Saliency Visualization of Notes - Example 3.	79
Figure 34. Mid-Fusion Gate Attention Distribution.....	81
Figure 35. Late Fusion Feature Importance (Structured MLP vs ClinicalBERT MLP Probabilities).	81

List of Tables

Table 1. Summary of Key Metrics and Model Performance in Readmission Prediction Studies.	22
Table 2. Summary of Validation Techniques in Readmission Prediction Studies.	24
Table 3. Summary of Model Performance and Key Findings in Readmission Prediction Studies.	25
Table 4. Description of MIMIC-IV Dataset Before Preprocessing.....	34
Table 5. Description of Dataset Post-Preprocessing and Feature Engineering.....	39
Table 6. Performance metrics for baseline machine learning models on structured data.	67
Table 7. Performance comparison of Structured MLP, ClinicalBERT MLP, Late Fusion, and Mid-Fusion models.	72
Table 8. Important Words per Example (Word Importance Table).	80
Table 9. Final Model Comparison Table (Structured MLP, ClinicalBERT MLP on Fine-Tuned Embeddings, ClinicalBERT Fine-Tuned, Late Fusion RF, Hybrid Fusion MLP)	82

Chapter 1 - Introduction

1.1 Introduction

Readmissions to hospitals less than 30 days after the time of discharge are of growing concern to the healthcare system across the world. Elderly people, usually defined as those aged 65 years and above, stand at an increased risk of readmission due to the presence of comorbidities and declining health. Continuous readmissions in elderly patient groups stretch healthcare systems to the breaking point, thereby causing an increase in the cost of the health system. This research aims to study in-depth the prediction analysis of the features and factors that contribute to the accuracy of prediction, through which 30-day readmission events are predicted among elderly patients. Data from the MIMIC-IV dataset (Johnson et al., 2023) is used to focus specifically on elderly patients and their unique challenges within the healthcare system. The study will include a comprehensive data preprocessing, preparation, and validation of models with the global aim of providing better care for the identified vulnerable population. The subject of this project will be elderly patients, who, by virtue of the medical, cognitive, and functional challenges they face, often require complex care coordination. Well-developed predictive models of elderly readmissions enable hospitals to coordinate their resource allocation and target interventions aimed at reducing readmission rates and improving patient outcomes better. The current models have low generalisation capabilities and are inefficient in learning from practical healthcare data due to missing data, unbalanced classes, and noisy data, which reduce their performance characteristics. In this research, these problems with data preprocessing are tackled, and advanced machine learning models are leveraged to improve the performance and interpretability of predictive models.

1.2 Research Question

This study aims to address a critical issue in healthcare analytics by exploring the factors influencing hospital readmissions and improving predictive performance through advanced machine learning techniques. The research investigates the following question:

"What are the key clinical and demographic factors accurately predicting 30-day hospital readmissions among elderly patients, and how can advanced machine learning models and

preprocessing techniques be leveraged to enhance the accuracy and interpretability of these predictions?"

1.3 Research Objectives

The following research objectives are designed to systematically address the prediction of 30-day hospital readmissions among elderly patients using machine learning models.

1. To develop a data pipeline for cleaning, handling missing values, mapping, scaling, and outlier removal, focusing on elderly patients.
2. To Build and evaluate machine learning models (logistic regression, random forests, XGBoost) to predict 30-day readmission in elderly patients.
3. To engineer features to capture key readmission factors, such as chronic conditions, medication use, and discharge details.
4. To assess model performance with metrics such as accuracy, precision, recall, F1-score, and AUROC, with special attention to class imbalance.
5. To implement and evaluate deep learning models and compare to traditional methods.
6. To ensure model interpretability using feature importance and Explainable AI (XAI) methods such as SHAP values to identify influential predictors for actionable healthcare insights.

1.4 Analysis Framework

Below is Figure 1, which is the analysis framework divided into six stages. In the first stage, a thorough background research and literature review will be completed, followed by the data acquisition for the analysis. Data preprocessing is next, followed by feature engineering and model development. In the final stage, the focus is on the evaluation and interpretation of the model performance and results. This structured approach supports the analysis of elderly patient readmissions.

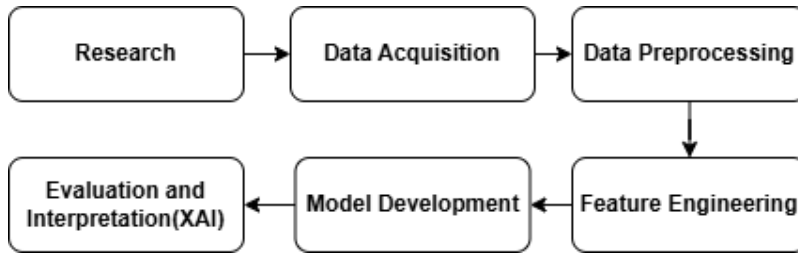


Figure 1. Analysis Framework.

1.5 Structure of Dissertation

This dissertation is structured into six chapters. Chapter 1 provides an opening to the dissertation. Chapter 2 entails a detailed literature review, discussing the impact and significance of hospital readmissions among elderly patients and common challenges associated with healthcare data. There is also a focus on prior studies tackling hospital readmissions. Chapter 3 outlines methodologies that include data acquisition, preprocessing, feature engineering, model development, and explainability techniques. Chapter 4 presents the results and findings of the analysis. Chapter 5 discusses limitations encountered and possibilities for future analysis and implementation. Chapter 6 ends this dissertation with the conclusion.

Chapter 2 - Literature Review

The current research on 30-day hospital readmission prediction is evaluated within this chapter. The studies that focused on elderly patients were evaluated. However, the limited number of studies about this population led to the inclusion of general 30-day readmission studies, focusing on traditional machine learning and deep learning methods on structured and unstructured electronic health record (EHR) data. The search strategy used PubMed and Cochrane databases to search for publications from the last 10 years that included terms such as “readmissions,” “hospital readmissions,” “30-day readmissions,” “Machine Learning,” and “predictive models.” The search yielded 571 articles, leading to a full review of 42 articles before selecting 28 for detailed analysis based on their methodological quality and study relevance.

2.1 Hospital Readmission Prediction

The quality of healthcare services and operational efficiency of hospitals depend heavily on 30-day readmission rates. Readmissions expose complex clinical patterns, revealing frailty,

multimorbidity, and socio-environmental vulnerabilities, particularly among elderly patients. Reducing readmissions serves two important purposes: helping both financial sustainability under CMS reimbursement policies and improving patient outcomes and quality of life (Zhao et al., 2020).

The broad implementation of EHR systems provides opportunities to forecast and stop hospital readmissions. Predictive models have traditionally relied on structured data elements, which include diagnoses, lab results, and patient demographics. Unstructured clinical narratives contain detailed patient information, exceeding the capabilities of structured data systems. However, the opportunities for prediction face significant obstacles because of missing data, inconsistent documentation, and built-in data biases.

Machine learning (ML) and deep learning have transformed readmission prediction. Traditional methods like logistic regression, decision trees, and Random Forests have been extended with more sophisticated architectures such as gradient boosting and transformers like ClinicalBERT (Wong et al., 2021). While these models show promise, their generalization to complex elderly populations remains imperfect. Elderly care involves managing multiple chronic conditions, medications, cognitive decline, and psychosocial factors, essentially elements that structured data alone often inadequately captures.

Ethical considerations further shape predictive modelling in healthcare. Ensuring model interpretability, data privacy, and fairness across diverse patient groups is essential. Predictions must be transparent to foster clinical trust and avoid exacerbating existing healthcare disparities.

This review critically explores modelling techniques, data preprocessing strategies, evaluation methodologies, and ethical challenges related to predicting 30-day elderly readmissions, setting the foundation for developing more sustainable, interpretable, and clinically deployable predictive models.

2.1.1 Key Clinical and Demographic Factors for 30-Day Readmissions

Numerous studies identify consistent predictors of 30-day readmissions, such as comorbidity burden (e.g., Charlson Comorbidity Index), abnormal lab values (e.g., sodium, albumin), prior hospitalisations, and socioeconomic status (Deschepper et al., 2019; Nguyen et al., 2021; Saleh

et al., 2020). Age and frailty further increased the risk among elderly patients. Integrating clinical and social health determinants is critical for developing robust predictive models.

2.2 Preprocessing in Predictive Modelling

Effective preprocessing is crucial to developing stable, generalisable, and accurate predictive models, particularly in healthcare, where data sets often contain missing values, imbalanced classes, and noisy data. These problems must be tackled carefully to ensure the validity and generalisability of models.

2.2.1 Handling Missing Data

Missing data is prevalent in EHRs, where incomplete fields are frequent. Basic techniques like mean and median imputation are simple but risk introducing bias in complex datasets (Nguyen et al., 2021; Brom et al., 2020). More advanced approaches, such as K-Nearest Neighbours (KNN) imputation, leverage patient similarity to improve accuracy (Aldhoayan & Khayat, 2022), although generalisability remains a limitation of these approaches. Multiple Imputation by Chained Equations (MICE) has shown robust performance (Huang et al., 2021; Saleh et al., 2020) but is computationally intensive and may increase bias against minority groups. Tailored imputation strategies focused on physiological parameters offer more scalable solutions in critical care settings (Loreto et al., 2020).

2.2.2 Feature Engineering and Selection

Identifying key predictors like Charlson Comorbidity Index (CCI), abnormal labs (e.g., sodium, albumin), and prior admissions have been critical for readmission models (Lo et al., 2021; Michailidis et al., 2022). However, CCI may be less predictive for acute conditions, emphasising the need for dynamic feature engineering. Dimensionality reduction techniques like Recursive Feature Elimination (RFE) and Principal Component Analysis (PCA) can simplify models (Zhou et al., 2016; Zhang et al., 2021), though biases in healthcare data can limit their effectiveness. Combining clinician expertise with ML-based feature selection improves fairness.

Interpretability tools such as SHAP and LIME enhance trust but come with computational costs (Carvalho et al., 2023; Mohanty et al., 2022). For unstructured data, models like BERTopic and ClinicalBERT have been successfully employed for interpretable feature extraction (Chiu et al., 2024).

2.2.3 Addressing Data Imbalance

Class imbalance remains a significant obstacle in readmission prediction. Although they generate synthetic samples from existing patterns, techniques like SMOTE have improved F1-scores and recall (Chiu et al., 2024; Loreto et al., 2020). Alternatives such as generative adversarial networks (GANs) or fairness-aware resampling may offer more diverse synthetic populations. Cost-Sensitive Learning, which penalises misclassifications of minority classes, has shown effectiveness when combined with ensemble models like Random Forest (Ashfaq et al., 2019; Brom et al., 2020). Hybrid strategies that blend oversampling and cost-sensitive approaches have further boosted performance, particularly in ICU readmission models (Loreto et al., 2020).

2.3 Machine Learning (ML) Models in Predictive Analytics.

Predictive analytics through machine learning (ML) has revolutionised the healthcare field. ML introduced innovative methods to forecast patients' 30-day hospital readmission risk. This section investigates the transition from traditional statistical approaches to modern, state-of-the-art ML techniques while discussing their applications, challenges, and advancements.

2.3.1 Traditional Predictive Methods

Traditional models such as Logistic Regression (LR) remain widely used in hospital readmission prediction due to their transparency and ease of interpretation, helping foster clinical trust. However, LR's reliance on linear assumptions limits its ability to model complex, non-linear patient relationships. For example, Low et al. (2015) compared LR to the LACE index for 30-day readmissions, achieving a slightly better AUC-ROC (0.65 vs. 0.63). Similarly, Nguyen et al. (2021) reported moderate performance for LR models (C-statistic 0.66) in a safety-net hospital, noting that hybrid approaches combining clinician judgment with EHR features improved both interpretability and predictive power.

The healthcare industry continues to use Logistic Regression (LR) for hospital readmission predictions due to its ability to provide clear, transparent results that clinicians can interpret. However, the linear assumptions in LR prevent the model from effectively capturing complex non-linear relationships between patients. The study by Low et al. (2015) demonstrated that LR produced a better AUC-ROC result (0.65 vs. 0.63) than the LACE index when predicting 30-day

readmissions. The C-statistic of LR models reached 0.66 according to Nguyen et al. (2021) in their study of a safety-net hospital, while hybrid approaches that combined EHR features with clinician judgment improved both interpretability and predictive power.

2.3.2 Modern Machine Learning Techniques

Current ML techniques overcome many of the restrictions of conventional methods and enable the modelling of non-linear relationships, interactions, and high-dimensional data. These methods include tree-based algorithms, ensemble models, deep learning, knowledge graph embeddings, and multi-model hybrid models.

2.3.3 Tree-Based Models

Tree-based models like Random Forest (RF) and Gradient Boosting Machines (GBM) are popular for their robustness and ability to model complex feature interactions. However, they risk overfitting in high-dimensional datasets and often require post-hoc explainability methods like SHAP. Huang et al. (2021) found RF models frequently outperformed alternatives for readmission prediction (AUC-ROC > 0.70). Deschepper et al. (2019) reported a 0.77 AUC-ROC using hierarchical ICD-10 features in RF models. Shi et al. (2023) achieved an internal AUC-ROC of 0.85 with GBM for ICU readmissions but observed poor external performance due to demographic shifts, highlighting a critical generalization challenge. Michailidis et al. (2022) showed RF models' adaptability to operational hospital data with an Area Under the Receiver Operating Characteristic Curve (AUROC) of 0.78.

2.3.4 Ensemble Methods

Ensemble methods like AdaBoost and CatBoost combine multiple models to improve prediction accuracy. Eckert et al. (2019) applied AdaBoost for military hospital readmissions, achieving AUC-ROC scores of 0.68 (retrospective) and 0.64 (prospective). Lo et al. (2021) used CatBoost to predict 14-day readmissions, reporting an impressive AUC-ROC of 0.99 and AUC-PR of 0.77. Mohanty et al. (2022) also endorsed ensemble methods, highlighting CatBoost and stacking classifiers as balancing accuracy and interpretability. However, ensembles demand high computational resources and can amplify biases by overfitting hard-to-classify cases. Future

directions must explore bias-mitigation techniques to ensure equitable performance across patient groups.

2.3.5 Deep Learning Models

Deep learning models effectively extract patterns from sequential and high-dimensional data. However, there are still many challenges that prevent real-world implementation. Firstly, deep learning models require extremely high-quality datasets, due to ethical challenges discussed in Section 2.7.4. De-identification and security measures are imposed, leading to incomplete datasets that can decrease the model's effectiveness. Secondly, Long Short-Term Memory (LSTM) networks are particularly suitable for temporal data such as EHRs. However, they need high computational power, making it difficult to implement them in resource-constrained hospitals. Lastly, deep learning models are known as "Black Boxes," due to their lack of interpretability; this lack of transparency limits their adoption by clinicians and researchers. Future research can develop tailored Explainable AI(XAI) techniques for healthcare models. Additionally, there should be a focus on privacy-preserving techniques, such as federated learning, which enables decentralised training without direct data sharing, maintaining patient confidentiality. Ashfaq et al. (2019) used LSTM networks to model the sequential patient data and achieved an AUC-ROC of 0.77 for CHF patients. When validated temporally, the performance increased to 0.83, illustrating that LSTM can capture a patient's history over time. Additionally, Chiu et al. (2024) employed BERTopic coupled with LSTM networks for the analysis of discharge summaries and clinical data where the model achieved an AUC-ROC of 0.7994.

2.3.6 Multi-Modal and Hybrid Models

The use of multiple data modalities such as structured, unstructured, and temporal has become essential in enhancing prediction accuracy. Sheetrit et al. (2023) used multi-modal models that incorporated ICD-9 codes, time series data, and discharge notes. However, combining this can increase model complexity and computational costs, the combination did not improve the performance of the models as they were trained solely on unstructured discharge notes alone. Moerschbacher and He (2023) demonstrated that the combination of structured and unstructured EHR data enhanced the prediction accuracy in the ICU readmissions, and Random Forest and

Logistic Regression models were the best in combined datasets. There are still prevalent challenges with these approaches. Enhancing data harmonization across hospitals is difficult due to EHR systems varying across different hospitals, making it difficult to standardise data format. Developing standardised frameworks that can integrate structured and unstructured EHR data is vital to improve multi-model models in clinical settings.

2.3.7 Model Performance Across Studies

The advancement in learning algorithms has revolutionised the evolution of machine learning models in the prediction of hospital readmissions. Tree-structured and ensemble methods are still popular for their accuracy and interpretability, although deep learning and knowledge graphs provide new opportunities to capture complex relationships and to enrich the data contexts. However, there are still challenges such as the inability and generalisation of models, computational costs, and the integration of multi-modal data, requiring further research and investigation.

2.4 Leveraging Electronic Health Record (EHR) Data

The use of electronic health records (EHRs) has revolutionised the application of predictive analytics in the healthcare industry thanks to the collection of abundant and integrated datasets that capture patient data across different hospital environments. EHRs contain both structured data such as demographic data, vital signs, and test results, and unstructured data in the form of clinical notes, which have proven to be a valuable source for building predictive models. This section explores the opportunities and challenges of applying predictive modelling with EHR data, with a focus on hospital readmissions.

2.4.1 Structured vs. Unstructured Data

EHR data can be categorised into structured and unstructured data. Structured data contains quantifiable information such as a patient's age, gender, comorbidities, medications, and test outcomes. Most traditional and machine learning models are developed using structured data due to its standardised format, making it more readily integrated into traditional statistical and ML models (Huang et al., 2021; Moerschbacher & He, 2023). Unstructured data includes clinical notes, discharge summaries, and imaging reports that lack rich contextual and qualitative insights

that structured data lacks. The application of Natural Language Processing (NLP) techniques enables the extraction of meaningful insights from complex unstructured data providing value enhancing the prediction accuracy (Sheetrit et al., 2023; Nguyen et al., 2016).

2.4.2 Key Predictors from EHR Data

Several common predictors of readmission risk have been identified in the reviewed studies. The CCI is a widely used measure which assesses the severity of comorbid conditions and has been found to be highly associated with hospital readmissions (Low et al., 2015; Nguyen et al., 2021). In addition, Vital signs, and lab results measures such as sodium, albumin, and vital sign instability at discharge have been identified as significant predictors (Lo et al., 2021; Carvalho et al., 2023; Aldhoayan & Khayat, 2022). Furthermore, utilization of metrics including the number of prior admissions and visits to the emergency department help to capture patterns and risks of the patient (Zhou et al., (n.d); Moerschbacher & He, 2023).

2.4.3 Natural Language Processing (NLP) for Unstructured Data

The application of NLP techniques has allowed the integration of unstructured EHR data in the development of predictive models. ClinicalBERT, an NLP model was used to analyse the discharge notes and outperformed traditional models in predicting ICU readmissions with an AUROC of 0.75 (Sheetrit et al., 2023; Mohanty et al., 2022). Named Entity Recognition (NER) was employed to extract specific entities like diagnoses and treatments from the clinical notes to create additional model features (Ashfaq et al., 2019; Moerschbacher & He, 2023). Sentiment analysis that determines the tone and sentiment in patient notes can also help to enhance the risk assessment with additional context such as psychosocial factors (Shi et al., 2023; Loutati et al., 2024).

Despite advancements, NLP models still face challenges in their application to hospitals.

Inconsistent clinical documentation or ambiguous abbreviations and details can cause noisy data in predictive models. Additionally, clinical judgement when taking notes for patients can cause bias if select patient groups are underrepresented or misclassified, affecting prediction fairness. Robust preprocessing techniques, bias mitigation strategies, and continuous model validation to ensure reliability for clinicians.

2.4.4 Challenges of EHR Data

Although rich in resources for predictive modelling, EHR data is not without its flaws. Missing, or partially complete, can lead to a decrease in the performance of the model. Select patient demographics such as underinsured populations or patients who receive fragmented healthcare will have incomplete records, causing biased predictions that may misrepresent the readmission risk. These gaps can be addressed through advanced imputation techniques such as MICE and domain-specific imputation (Nguyen et al., 2021; Aldhoayan & Khayat, 2022). The combination of structured and unstructured data is complex and leads to increased computational overhead (Lo et al., 2021; Moerschbacher & He, 2023). The use of sensitive patient information mandate strict adherence to data protection and security protocols (Zhou et al., (n.d); Talwar et al., 2023). Due to ethical challenges discussed in Section 2.7.4, privacy preserving ML techniques such as federated learning offer a potential solution.

2.5 Model Evaluation

Machine learning (ML) models in healthcare need to be evaluated to determine their validity, efficiency, and readiness to use in clinical practice. For 30-day hospital readmissions, robust evaluation strategies involve a combination of performance metrics, interpretability tools, and validation procedures. This section synthesises findings from various studies, providing a comprehensive perspective.

2.5.1 Key Metrics for Evaluation

Table 1. Summary of Key Metrics and Model Performance in Readmission Prediction Studies.

Study	Model	Key Metric	Performance
Deschepper et al. (2019)	Random Forest	AUC-ROC	0.77 (Diagnosis-level predictors)
Ashfaq et al. (2019)	LSTM	AUC-ROC	0.83 (Temporal validation)
Eckert et al. (2019)	AdaBoost	AUC-ROC	0.68 (Outperformed decision trees & logistic regression)
Loreto et al. (2020)	Random Forest	AUC-ROC	0.91 (ICU readmission prediction)
Nguyen et al. (2021)	Hybrid Model (Clinician + EHR)	Precision & Recall	Precision: 78.1%, Recall: 75.2%
Lo et al. (2021)	CatBoost	Precision & Recall	Precision: 94.7%, Recall: 56.0%
Michailidis et al. (2022)	Balanced Random Forest	Sensitivity & AUC-ROC	Sensitivity: 0.70, AUC-ROC: 0.78
Shi et al. (2023)	GBM	F1-Score	0.43 (Tuned for balanced class predictions)
Carvalho et al. (2023)	Random Forest + Knowledge Graph	AUC-PR	0.69 (Focused on minority cases)

Table 1 illustrates the performance metrics that are essential for assessing the predictive ability and utility of ML models. Studies adopt a variety of metrics depending on clinical and computational priorities. AUC-ROC (Area Under the Receiver Operating Characteristic Curve) metric measures a model's ability to discriminate between positive and negative outcomes. A Long-Term Short Memory (LSTM) network was employed by Ashfaq et al. (2019), with temporal validation of the model. However, AUC-ROC may be misleading in imbalanced datasets. In the retrospective validation, Eckert et al. (2019) reported that AdaBoost outperformed decision trees and logistic regression. In the case of Loreto et al. (2020), the AUC-ROC of the Random Forest model predicting ICU readmissions indicated that tree-based models are very efficient. The model's accuracy and positive rate are better for correct positive predictions when the data is imbalanced. The best values of precision and recall were obtained when Nguyen et al. (2021) proposed a hybrid model that combines the expertise of clinicians with the output of EHR-based predictions. Lo et al. (2021) obtained high precision with CatBoost models for short-term readmission prediction, but the recall was moderate because of data imbalance. Michailidis et al. (2022) employed Balanced Random Forests. The F1-Score is the harmonic mean between sensitivity and specificity. Shi et al. (2023)'s GBM models were tuned to improve their model's ability to make balanced class predictions in the case of ICU

readmissions. The AUC-PR is a metric that focuses more on the minority classes, and since many datasets are imbalanced, this metric is useful. Carvalho et al. (2023) presented Random Forest models that were enhanced by knowledge graph embeddings to depict the potential of the model in identifying the minority cases.

2.5.2 Model Interpretability

Interpretability is a cornerstone of model adoption in clinical settings, providing clinicians with actionable insights. As a feature importance tool, SHAP (Shapley Additive Explanations) is widely used for feature importance analysis. SHAP quantifies each feature's contribution in each prediction. Utilizing SHAP, Lo et al. (2021) revealed that CatBoost models main key explanatory features included discharge medications and lab results, thus increasing the model's clinical relevance. Mohanty et al. (2022) employed SHAP to explain the influence of frailty indicators in readmission predictions, facilitating targeted interventions. A commonly used technique for explaining an individual prediction is LIME (Local Interpretable Model-agnostic Explanations). Nguyen et al. (2021) applied LIME to explain the discrepancies between the predictions of clinicians and ML models, to enhance the explainability of the decision-making process. However, LIME only explains predictions locally and does not provide a global understanding of how the model makes decisions across the dataset. Feature importance scores like tree-based models inherently provide feature importance rankings. Deschepper et al. (2019) implemented Random Forest feature importance to identify ICD-10 diagnosis codes as the most important predictors of readmissions. Brom et al. (2020) highlighted CART's ability to generate interpretable paths for nurse-led interventions in safety net hospitals. Future research should explore hybrid interpretability techniques, combining SHAP, LIME, and clinician knowledge to provide accurate, meaningful explanations.

2.5.3 Validation Techniques

Table 2. Summary of Validation Techniques in Readmission Prediction Studies.

Study	Validation Type	Key Findings
Mesgarpour et al. (2017)	Internal Validation (Cross-Validation)	Ensemble Bayesian models achieved consistent AUC-ROC across folds.
Michailidis et al. (2022)	Internal Validation (k-Fold Cross-Validation)	k-Fold cross-validation ensured balanced performance metrics.
Shi et al. (2023)	External Validation	AUC-ROC declined from 0.85 to 0.60 due to demographic variability and dataset shifts.
Moerschbacher & He (2023)	External Validation	Random Forest models achieved AUROC ~ 0.75 on combined datasets.
Ashfaq et al. (2019)	Temporal Validation	Temporal validation improved LSTM model AUC-ROC from 0.77 to 0.83, showing adaptability to evolving patient trends.

Validation ensures that predictive models generalise beyond their training datasets, highlighting their robustness in diverse settings. Internal validation techniques, such as cross-validation, are commonly employed for model evaluation within the same dataset. External validation evaluates models by applying them to separate datasets, assessing generalisability. Table 2 outlines Shi et al. (2023), noting a significant AUC-ROC decline during external validation, attributing this to demographic variability, dataset shifts, and institutional differences. These results underscore model transferability to other hospitals as models trained on specific hospital EHR datasets may not generalise well to other hospital EHR datasets due to differences in documentation, treatments, or health disparities. Temporal Validation of models on temporally distinct datasets mimics real-world deployment. Future work that incorporates cross-hospital validation frameworks is vital to tackle the generalisability issues of models.

2.5.4 Model Performance

Table 3. Summary of Model Performance and Key Findings in Readmission Prediction Studies.

Study	Focus Area	Key Findings
Nguyen et al. (2021)	Handling Missing Data, Model Evaluation	Utilized KNN imputation; evaluated hybrid ML models combining clinician judgment and EHR data.
Brom et al. (2020)	Readmission Rates	30-day readmission rate of 11.2% (n=242); discussed cost-sensitive methods for imbalanced datasets.
Aldhoayan & Khayat (2022)	Imputation Techniques, Predictive Modeling	Evaluated various ML models for seven-day readmissions; CatBoost achieved highest AUC.
Huang et al. (2021)	Iterative Imputation, Model Evaluation	Advocated for MICE; emphasized fairness-aware ML techniques to reduce algorithmic biases.
Saleh et al. (2020)	Iterative Imputation, Robust Predictive Models	Reviewed iterative techniques for healthcare data; observed improvements in prediction robustness.
Loreto et al. (2020)	Imputation, SMOTE	Highlighted hybrid imputation and oversampling strategies for ICU readmissions.
Lo et al. (2021)	Clinical Predictors, SHAP	Identified discharge medications and lab results as critical features in CatBoost models.
Michailidis et al. (2022)	Random Forest, SMOTE	Balanced Random Forest showed improved sensitivity for readmission prediction.
Carvalho et al. (2023)	Knowledge Graphs, Explainability	Developed RDF2Vec embeddings, improving ICU readmission model accuracy and interpretability.
Chiu et al. (2024)	BERTopic, NLP	Combined LSTM and BERTopic for integrating structured and unstructured data.
Ashfaq et al. (2019)	LSTM, Temporal Validation	Improved CHF readmissions model accuracy with LSTM; emphasized importance of temporal validation.
Low et al. (2015)	Logistic Regression	Compared LACE index with logistic regression, showing better discrimination by the latter.
Deschepper et al. (2019)	Random Forest, Feature Importance	Used Random Forest with ICD-10 codes for readmission prediction.
Shi et al. (2023)	Gradient Boosting Machines (GBM)	Applied GBM for ICU readmissions, showing internal validation improvements.
Eckert et al. (2019)	AdaBoost, Workflow Integration	Evaluated AdaBoost for predicting military hospital readmissions.
Meng et al. (2023)	Explainable Boosting Machines (EBMs)	Discussed the use of EBMs for interpretable ICU readmission prediction models.
Sheetrit et al. (2023)	Multi-Modal Models, ClinicalBERT	Integrated structured/unstructured EHR data; NLP showed notable improvement.
Moerschbacher & He (2023)	EHR Data Integration	Highlighted combining discharge summaries and structured EHR data for ICU readmissions.
Talwar et al. (2023)	Ethical Considerations, Privacy Regulations	Emphasized compliance with GDPR and HIPAA; highlighted risks of algorithmic bias.
Lin et al. (2019)	LSTM, Sequential Modeling	Applied LSTM for temporal modeling in readmission prediction.

2.5.5 Challenges in Evaluation

Despite advancements, there are still many challenges in evaluating ML models. Metrics such as accuracy, are skewed by class imbalance, emphasizing the need for precision, recall, and F-Score. Table 3 outlines the key findings of previous models and focus areas of related literature. In addition to utilizing KNN imputation, Nguyen et al. (2021) implemented SMOTE and cost-sensitive algorithms to mitigate class imbalance. Complex models often excel in internal validation but fail to generalise. Sheetrit et al. (2023) reported overfitting issues when testing GRU RNN models on smaller datasets. Evaluating models for biases and ensuring compliance with privacy regulations, such as GDPR and HIPAA, are essential for deployment to ensure models are clinically effective and ethically responsible.

2.6 Challenges and Limitations

Despite recent improvements in the development of models for predicting 30-day hospital readmissions, there are still several factors that hinder their effectiveness and applicability. These challenges result from complex data, model limitations, and operational barriers, which need to be resolved for real-world implementation to be dependable.

2.6.1 Data Challenges

Electronic Health Record (EHR) datasets often suffer from missing values, imbalanced classes, and integration complexities. To address missingness, Nguyen et al. (2021) applied KNN imputation, while Huang et al. (2021) showed that advanced iterative techniques improved model robustness, particularly for Random Forests. Wang and Zhu (2022) proposed hybrid imputation strategies for high-dimensional datasets.

Class imbalance remains a significant barrier, biasing metrics like AUC-ROC. Shi et al. (2023) observed degraded performance in minority classes without oversampling, whereas Ashfaq et al. (2019) and Loreto et al. (2020) demonstrated that techniques like class-weighting and cost-sensitive SMOTE improved recall for readmissions.

Integrating structured and unstructured data further introduces preprocessing and computational complexity. Sheetrit et al. (2023) found that while fusion modestly enhanced performance, it also increased resource demands. Moerschbacher and He (2023) confirmed that combining discharge

summaries with structured EHR features significantly boosted ICU readmission prediction accuracy.

2.6.2 Model Challenges

The limitations that come with machine learning models further prevent their adoption by researchers. Most models have proven to be accurate with internal validation but fail to generalise across the different datasets, which is a massive constraint in healthcare implementation. Shi et al. (2023) reported that the performance of a GBM model, when trained on MIMIC-IV, decreased by a large margin on external datasets (AUC-ROC from 0.85 to 0.60). Carvalho et al. (2023) implied that although ensemble methods were accurate, they required careful hyperparameter tuning if overfitting is to be prevented while maintaining interpretability.

Deep learning models are known as black box models, due to their lack of transparency. Lo et al. (2021) demonstrated that the use of SHAP values proved to be useful in explaining the CatBoost models when predicting health outcomes, although it did increase computational costs. Chiu et al. (2024) employed the BERTopic model to provide explorable predictions from unstructured text data, this integrated the best aspects of both topic modelling and deep learning. Nguyen et al. (2021) stated that logistic regression was the preferred model in clinical practice due to the model simplicity, even with the model's low accuracy. The LSTM and GRU RNNs require significant computational power to be implemented, this limits their potential scalability and feasibility in hospitals with limited resources. Ashfaq et al. (2019) explained that their LSTM model was trained on large EHR data sets, and this required the use of GPU acceleration to manage this load efficiently. Wang and Zhu (2022) pointed out that the use of ensemble methods like Gradient Boosting is computationally expensive when applied to multi-modal data.

2.6.3 Operational Challenges

Integrating predictive models into clinical workflows remains challenging. Low et al. (2015) emphasised the need for early prediction availability to allow timely clinician action, while Eckert et al. (2019) noted that models requiring complex preprocessing were often rejected by hospital staff due to operational burdens.

Privacy and ethical concerns are central. Shi et al. (2023) stressed adherence to GDPR and HIPAA regulations for handling sensitive EHR data. Algorithmic bias remains a major risk; Huang et al. (2021) and Moerschbacher and He (2023) highlighted how demographic disparities in EHR data can perpetuate inequalities if not carefully addressed.

Clinician trust is another barrier. Nguyen et al. (2021) found that hybrid models combining machine learning outputs with clinical expertise were more likely to gain acceptance. Future work should prioritise user-friendly, transparent decision-support systems that integrate smoothly into clinical workflows.

2.7 Research Gaps

The use of machine learning (ML) in predicting 30-day hospital readmissions has progressed a long way. Nevertheless, there are still currently various gaps in the current literature. This section discusses the latest trends and directions based on the findings of the recent studies reviewed.

2.7.1 Enhancing Data Quality and Accessibility

The accuracy of the predictive models is dependent on the quality and the comprehensiveness of the data used. Challenges such as addressing missing data, imbalanced datasets, and inconsistent EHRs can be addressed to enhance the performance of the models and their generalisability. While methods such as KNN and iterative imputation have proven to be quite effective, it is worthwhile exploring the possibility of implementing deep learning-based imputation methods to deal with high-dimensional data sets that have extensive missing values. Huang et al. (2021) and Zhang et al. (2021) emphasised the need for advanced imputation techniques to improve the reliability of models in the context of healthcare practice. Expanding the data set to include information from several institutions can help increase the generalisability of the study. Shi et al. (2023) and Aldhoayan & Khayat (2022) pointed out the necessity for collaborative efforts to gather large datasets and share them for the purpose of validation across geographic and demographic factors. Some of the variables that are not represented enough in the current models include socioeconomic status, housing stability, and caregiver support. Studies by Nguyen et al. (2021) and Meng et al. (2022) recommended the use of EHRs in conjunction with community

and public health information to address this gap and enhance the relevance of the predictive analysis.

2.7.2 Advancing Model Development

The development of new innovative algorithms and frameworks in the field of ML can enhance predictive models. The development of interpretable models or the integration of advanced XAI tools can gain the trust of clinicians when utilizing models. Tools such as SHAP and ClinicalBERT have been applied to bridge the complexity-awareness gap by providing transparent explanations of ML predictions, which was identified by Lo et al. (2021) and Hegselmann et al. (n.d.). While SHAP applies feature importance analysis, ClinicalBERT can enhance the understanding of unstructured data such as clinical notes. Combining statistical techniques in conjunction with ML algorithms is effective. Nguyen et al. (2021) recommended that the models should incorporate the unstructured input of the clinicians together with the ML predictions to improve precision, recall, and other metrics, respectively. One of the solutions that has addressed the concerns of data sharing already in this review of literature is federated learning, a privacy-preserving which trains models from decentralised data, ensuring compliance with HIPAA and GDPR regulations, as suggested by Huang et al. (2021) and Carvalho et al. (2023).

2.7.3 Expanding Data Modalities

In addition to improving model interpretability, another promising avenue is expanding the modalities of data inputs used for prediction. The application of models in healthcare will be enhanced by the inclusion of diverse unstructured sources of information in addition to the conventional structured EHRs. As mentioned in section 2.4.3, it is apparent that using NLP techniques such as ClinicalBERT could enhance predictions for ICU readmissions, improving the recurring issue of generalization. Information obtained from wearable devices is a potential source of information that has not been fully explored in general predictive modelling. Future research should utilise multi-model fusion techniques that combine wearable devices and NLP techniques with EHR datasets to improve personalised patient monitoring and readmission predictions.

2.7.4 Addressing Operational and Ethical Challenges

Successful adoption of ML models in healthcare requires addressing operational and ethical barriers. Models must deliver actionable insights that integrate easily into clinical workflows. Eckert et al. (2019) highlighted the need for real-time prediction systems to support discharge planning. Mitigating bias is critical. Huang et al. (2021) advocated for fairness-aware ML techniques, such as reweighting, adversarial debiasing, and counterfactual fairness, to reduce demographic biases in predictions. Adherence to privacy regulations like HIPAA and GDPR remains essential; Shi et al. (2023) and Michailidis et al. (2022) emphasised the need for transparency and strict compliance with ethical standards during model development and deployment.

2.7.5 Rigorous Validation and Benchmarking

Many studies highlight the need for rigorous validation frameworks to ensure real-world applicability and generalisability. Carvalho et al. (2023) proposed creating standard datasets and benchmarks for 30 days readmission models to enable constant comparison of different studies. This would help researchers improve generalization in models. Ashfaq et al. (2019) and Meng et al. (n.d.) evaluated the effectiveness of temporal validation and proved how it aided in enhancing the reliability of the models developed.

2.8 Conclusion

This review of the literature explores various strategies for predicting hospital readmissions with a key focus on machine learning methods employed on electronic health record (EHR) data. While significant progress and advancement has been made in predictive modelling, several important critical gaps remain, particularly in studies focusing on the elderly patients who are characterised by complex comorbidities, frailty, and polypharmacy. Despite the usefulness of general studies on readmissions and EHR based approaches, there is a lack of focused research investigating the needs and characteristics of elderly patients. Key challenges identified from the review include data pre-processing, feature selection, model testing and validation, model interpretability, practicality and feasibility, and ethical concerns. A common recurring theme of the reviewed studies was the need for rigorous validation and testing procedures, focusing on incorporating structured and unstructured data to enhance model performance and

generalisability. Due to the limited number of studies specifically focusing on elderly readmissions, this research seeks to address this gap by developing explainable, clinically relevant advanced predictive readmission models, for this high-risk cohort, leveraging deep learning's untapped potential. By advancing interpretable ML frameworks, this work aims to improve patient outcomes and optimise healthcare resource allocation, aligning with SDG goal 12 (Responsible Consumption and Production) and SDG 3 goal (Good Health and Well-being).

Chapter 3 - Methodology

This study employs a structured approach to develop predictive models for 30-day hospital readmissions in elderly patients. The methodology involves several key stages, including data retrieval and preprocessing, feature engineering, model development and evaluation.

Preprocessing steps were applied to individual datasets before merging to ensure data was consistent and complete. Feature engineering methods were employed to define interpretable features such as count-based columns for lab results and prescriptions. The data was filtered to only focus on elderly patients (aged 65+) to match the objectives of the study. Machine Learning models, including Logistic regression, random forests and deep learning architectures were implemented to predict the risk of readmission, with performance assessed using metrics such as accuracy, precision, recall, F1 score and AUC-ROC. To enhance model fairness and performance in handling class imbalance, techniques like SMOTE were applied while explainable AI methods such as SHAP values were employed to improve interpretability. This methodology achieves a balance between predictive accuracy and clinical relevance, providing valuable insights for effective healthcare decision-making.

A high-level overview of the flow of methods implemented can be found below. This includes data analysis, code implementation, and data preparation for individual datasets and merged datasets, and describes the machine learning algorithms used to reach the readmission prediction results.

3.1 Data Acquisition

For this study, the data was obtained from the MIMIC-IV database which is a public access electronic health record (EHR) dataset hosted on PhysioNet. To access the dataset, it is required to complete a PhysioNet credentialing process that entailed training modules and accepting the data use agreement. Due to the dataset is being exceptionally large, containing more than 10 million rows with both structured and unstructured data, it was important to use efficient retrieval strategies. The key tables used in the analysis include Admissions for patient visit information, Patients for demographic data, Diagnoses_ICD and Procedures_ICD for clinical coding data, Lab Events and Lab Items for laboratory results, Prescriptions for medication records, and MIMIC-IV Note for unstructured clinical notes including discharge summaries. As

some of the tables, for example lab_events were exceptionally large, Google Colab Pro with high RAM resources was used for data retrieval, and chunking methods were employed to help with memory management. Some preprocessing steps that were completed on the individual datasets included handling missing values, converting ICD-10 to ICD-9 codes, and extracting features such as lab result counts before merging them into a filtered dataset containing only data for elderly patients (65+). All the data handling was completed in accordance with HIPAA regulations and PhysioNet compliance standards to preserve data security and patient confidentiality.

3.1.1 Structured Dataset Overview

The analysis utilised core structured components from MIMIC-IV, focusing on admissions, diagnoses, procedures, prescriptions, and laboratory records. The Admissions table served as the central reference, providing key metadata such as admission timing, type, and patient discharge details. Demographic attributes like insurance type, marital status, language, and race were drawn from both the Admissions and Patients tables, with anchor_age used to compute age at the time of admission.

Diagnostic and procedural information was extracted from ICD-coded tables. These entries, identified by icd_code and icd_version, enabled construction of binary and count-based features reflecting the clinical context of each hospital stay. Similarly, the Prescriptions dataset captured medication orders, dosage routes, and durations, allowing derivation of features such as drug counts and indicators of polypharmacy exposure. Laboratory results, including raw values and abnormality flags, were sourced from Lab Events, with supplementary metadata retrieved via Lab Items. These tables enabled the aggregation of critical test results per admission. Table 4 summarises the structured datasets used in the study, including the number of raw columns and a brief description of each table's contents, to provide a concise reference.

Table 4. Description of MIMIC-IV Dataset Before Preprocessing

Dataset	Description	Number of Columns
Admissions	Admission metadata (times, location, insurance, etc.)	15
Patients	Demographics and anchor age/year	6
Diagnoses ICD	ICD diagnosis codes linked to admissions	5
Procedures ICD	ICD procedure codes with sequence and chartdate	6
Prescriptions	Medication prescriptions and administration details	24
Lab Events	Raw lab test results per admission	15
Lab Items	Lab item descriptions, fluids, and categories	3

3.2 Data Preprocessing

Before modelling, it was necessary to address several problems that were present in the raw MIMIC-IV data. These issues made pre-processing a crucial step to guarantee the quality, consistency, and relevance of the data for predictive modelling. The key issues identified included:

- Missing or incomplete data: A high percentage of clinical variables had missing entries, especially lab values and demographic details that required appropriate imputation techniques.
- Mixed coding systems: Diagnoses and procedures were coded in ICD-9 and ICD-10, thus necessitating the use of GEM files to convert the data into a consistent unified format.
- Large-scale datasets: Tables like lab_events and note contained millions of rows, requiring efficient data handling techniques such as chunking and batch processing.
- Inconsistent time formats: Time stamps were inconsistent throughout the tables and required standardization for chronological sorting and feature engineering.
- Multiple rows per admission: Several tables (e.g. labs, diagnoses, prescriptions) were in long format with multiple rows per admission, which needed to be aggregated into single row per admission data suitable for modelling.
- Class imbalance: Readmitted patients were underrepresented in the dataset, so techniques such as SMOTEENN (Batista, Prati, & Monard, 2004) for resampling were implemented to balance the target variable.
- Noisy and unstructured text data: Discharge summaries needed to be cleaned and transformed before ClinicalBERT could be used to process the data. The model was first

assessed in its pretrained form to extract embeddings, but to improve performance, it was fine-tuned with the dataset on the downstream task of 30-day readmission to enhance its ability to capture relevant context-specific language patterns. This approach highlighted the importance and benefits of fine-tuning.

- Defining the target variable: One of the key preprocessing steps was creating the binary label for 30-day readmission. This was determined by calculating the time gap between a patient's discharge and subsequent admission. If the next admission occurred within 30 days of discharge, the index admission was labelled 1 (readmitted); otherwise, it was labelled 0 (not readmitted).

Addressing these issues was crucial to optimising the dataset for modelling while maintaining clinical relevance and interpretability. Preprocessing steps were designed to ensure the data would be generalisable to real-world clinical settings and interpretable from both a clinical and modelling perspective. Structured features such as counts, summary statistics and temporal patterns were engineered to balance the predictive power with transparency. Figure 2 outlines the workflow of the preprocessing pipeline and subsequent sections describe each preprocessing step in detail, including data cleaning, transformation, feature engineering, class balancing, and fine tuning of ClinicalBERT.

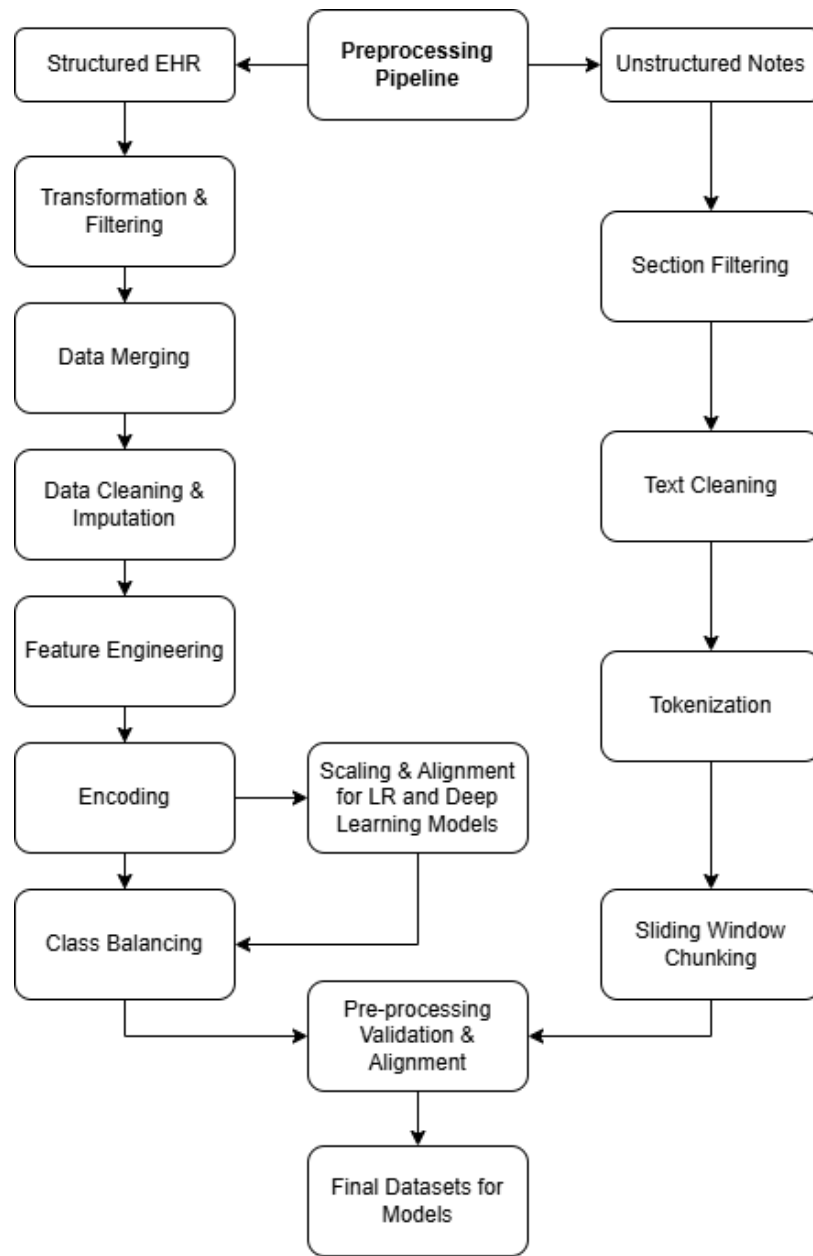


Figure 2. Overview of the data preprocessing pipeline for structured EHR features and unstructured discharge summaries.

3.2.1 Handling of Large-Scale Datasets NB

Due to the MIMIC-IV dataset's large volume, chunked extraction and processing were performed in Google Colab to avoid memory overload. Raw files were unzipped in parts and saved in Parquet format for efficient storage and I/O performance. Intermediate datasets (e.g.,

ICD mapping, lab aggregation, CLS embedding) were saved after key stages to support reproducibility and reruns.

3.2.2 Data Transformation and Data Filtering

Data transformation and filtering steps were applied to prepare for modelling. Diagnoses and procedures were harmonised across ICD-9 and ICD-10 codes using GEM files to ensure consistency during mapping. Timestamps were standardised to support chronological feature engineering. The study focused solely on elderly patients (65+), so early filtering by age reduced computational load and ensured cohort relevance. These steps ensured consistent, high-quality data aligned with study objectives.

3.2.3 Data Merging

The merging process began with patients and admissions data as the core identifiers, onto which labs, diagnoses, procedures, and prescriptions were joined using subject and admission IDs. To construct a modelling-ready dataset from the raw MIMIC-IV tables, data was initially merged from all relevant sources in long format, resulting in over 10 million rows due to one-to-many relationships in diagnoses and procedures. This format allowed for effective extraction of temporal features and count-based metrics at the row level, such as recent visit frequencies or diversity of codes. Once these features were computed, we aggregated the diagnoses and procedures per hospital admission (`hadm_id`) using a custom join function to retain unique, interpretable code sets. Duplicate `hadm_ids` were explicitly checked to ensure a strict one-row-per-admission structure before modelling. The resulting dataset, reduced to just over 200,000 rows, maintained one row per admission while preserving the richness of the original data. This aggregation was performed prior to imputation, exploratory data analysis, and modelling to ensure consistency, completeness, and scalability.

3.2.4 Data Cleaning and Imputation

Extensive cleaning and imputation were performed on the MIMIC-IV dataset to ensure modelling reliability. Missing values were addressed using imputation strategies tailored to variable types and clinical context. For categorical variables (e.g., admission type, race), mode imputation was used to retain dominant trends. Variables where absence may be informative

(e.g., language, marital status) were encoded with an "Unknown" label. Missingness indicators were added to directly allow the model to learn from absence patterns. Numerical variables were treated based on their semantics. Median imputation was applied to continuous features prone to outliers (e.g., lab values, length of stay). In contrast, zero imputation was reserved for counts where zero carries real clinical meaning (e.g., medication totals). More computationally intensive imputation methods (e.g., MICE, KNN) were deliberately avoided to preserve processing efficiency without sacrificing data integrity. Time formats and identifiers were standardised to maintain compatibility across datasets. This included unifying timestamp formats and resolving discrepancies in ID fields. These strategies enhanced dataset completeness, minimised bias, and maintained interpretability, laying a solid foundation for robust feature engineering and model development.

3.2.5 Feature Engineering

Feature engineering enhanced the dataset's predictive potential by creating clinically meaningful features informed by domain knowledge and data insights. This study strongly emphasised engineering simple, effective, and generalisable features. Count-based features summarised key events such as abnormal labs and prescription counts, capturing trends like clinical deterioration and polypharmacy linked to readmission risk. Additional lab-based statistics (mean, min, max, abnormal proportions) were generated to capture clinical instability.

Temporal features captured recent patient histories, including the number of admissions, diagnoses, procedures, and prescriptions in the past 30, 60, and 90 days. Other engineered variables included days since the last discharge, same-day readmission flags, care gaps, and length of stay. Together, these features provided comprehensive insights into clinical and behavioural factors associated with readmissions, supporting robust and interpretable model development. Table 5 below summarises the final engineered feature groups used in the structured dataset, including representative examples and corresponding feature counts.

Table 5. Description of Dataset Post-Preprocessing and Feature Engineering

Feature Group	Example Features	Count
Identifiers	hadm_id, subject_id	2
Admission Info	admittime, disctime, admission_type, admission_location, discharge_location	5
Demographics	insurance, language, marital_status, race, anchor_age, age_bin, age_bin_encoded	7
Lab Features	mean_lab_value, min_lab_value, max_lab_value, total_lab_count, unique_lab_count, most_common_lab_label, proportion_abnormal	7
Prescription Features	total_prescriptions, unique_drug_count, total_doses, zero_dose_proportion, polypharmacy_flag	5
Readmission Target	readmitted_30d	1
Temporal/History Features	days_since_last_discharge, num_visits_last_90_days, same_day_readmission_flag, gap_flag, length_of_stay	5
Diagnoses (Recent Counts)	num_diagnoses_last_30_days, num_diagnoses_last_60_days, num_diagnoses_last_90_days	3
Prescriptions (Recent Counts)	num_prescriptions_last_30_days, num_prescriptions_last_60_days, num_prescriptions_last_90_days	3
Procedures (Recent Counts)	num_procedures_last_30_days, num_procedures_last_60_days, num_procedures_last_90_days	3
Diagnosis/Procedure Diversity	num_unique_diagnoses, num_unique_procedures	2
Mapped Code Sets	mapped_diagnoses_icd9, mapped_procedures_icd9	2
Total Features		46

3.2.6 Encoding of Categorical and Code-Based Variables

To prepare the dataset for modelling, a combination of frequency encoding and one-hot encoding was applied. For high-cardinality clinical code features such as diagnoses, procedures, and lab test names, frequency encoding was used after exploding the space-separated values. This approach captured the importance of common clinical codes while reducing dimensionality. Additionally, the top 5 flags were created for each code type to mark the presence of dominant medical concepts. For low- to moderate-cardinality categorical features such as admission type, insurance, and language, rare category grouping was first performed (threshold: <100 instances), and then one-hot encoding was applied with `drop_first=True` to reduce multicollinearity for linear models. These steps ensured that categorical data were efficiently encoded, clinically interpretable, and compatible with both traditional and deep learning models.

3.2.7 Final Scaling and Alignment

To prepare the structured data for modelling algorithms sensitive to feature scales, such as Logistic Regression and neural networks, appropriate scaling techniques were applied. A mix of RobustScaler, MinMaxScaler, and StandardScaler was used based on the distribution of each feature group. Lab-based summary statistics (e.g., mean, min, max lab values) were scaled using RobustScaler due to their susceptibility to outliers. Temporal count features reflecting diagnoses, procedures, and prescriptions in recent time windows were normalised using MinMaxScaler, while age was standardised using StandardScaler. The scaled dataset was saved separately in Parquet format for use in linear and deep learning models. Additionally, prior to modelling, all structured records were aligned with the `hadm_ids` present in the notes dataset to ensure fairness and consistency in hybrid and text-based modelling workflows. This ensured that each admission used for training or evaluation was represented uniformly across both modalities.

3.2.8 Text Data Processing

Unstructured discharge summaries were pre-processed to enhance readmission prediction. Notes were downloaded in compressed format and chunked in Colab to avoid memory overload. Only `subject_id`, `hadm_id`, and text columns were retained, excluding empty notes. Clinical section-filtering using regex extracted key sections (e.g., HPI, hospital course, discharge instructions). Texts were cleaned, lowercased, and tokenized into overlapping chunks (size 384, stride 256), using a sliding window strategy to preserve full context. Chunks were encoded with the ClinicalBERT tokenizer and fine-tuned for 30-day readmission prediction. Attention pooling combined chunk-level [CLS] embeddings into admission-level representations, enriching the dataset with contextual patient insights.

3.2.9 Class Balancing

The dataset exhibited a notable class imbalance, with only ~19% of admissions resulting in 30-day readmission. To address this, the hybrid class balancing method SMOTEENN was implemented. SMOTE (Synthetic Minority Over-sampling Technique) generated new synthetic samples for the minority class, while ENN (Edited Nearest Neighbours) removed noisy majority instances, improving overall class separation and reducing overfitting risk. Due to the dataset's enormous size, batch-wise resampling was used to apply SMOTEENN in manageable chunks,

reducing memory load and ensuring efficiency. Sampling_strategy of 0.4 and 0.5 was selected to achieve better recall without excessively lowering precision, which is crucial for minimizing false negatives in clinical prediction.

This balancing technique was complemented by Focal Loss (Lin et al., 2017) during model training. Focal Loss down-weighted well-classified (majority) examples and emphasised harder-to-classify (minority) cases, making it ideal for imbalanced medical datasets. It was particularly effective for deep learning models like the MLP and ClinicalBERT, enhancing their sensitivity to readmitted patients. By combining SMOTEENN for data-level balancing with Focal Loss for model-level emphasis, this dual approach improved recall and robustness, allowing the models to better detect patients at risk of readmission.

3.2.10 Cross-Cutting Preprocessing Strategies

Several cross-cutting strategies enhanced efficiency, fairness, and interpretability. Memory optimization involved down casting data types and saving in Parquet format. Label leakage was prevented by excluding future-outcome features. Structured and unstructured datasets were aligned via hadm_id for multi-modal modelling. Explainability was supported through SHAP values for structured models and attention pooling for ClinicalBERT embeddings.

3.3 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was conducted after initial preprocessing (cleaning, imputation, filtering, and feature engineering) to uncover patterns and assess the dataset's structure. Analyses focused on demographic characteristics, admission trends, clinical measurements, and potential predictors of 30-day readmission. This stage guided preprocessing decisions and ensured the dataset was well-understood before applying synthetic sampling or merging of embeddings.

3.3.1 Patient Demographics and Cohort Overview

The cohort was filtered to patients aged 65 and above, aligning with the study's focus on elderly readmission risk. Most patients were between 65 and 75 years old, with a spike at age 90 due to MIMIC-IV's age truncation policy (Figure 3). The observed 30-day readmission rate was 19%,

highlighting a class imbalance that informed later resampling strategies (Figure 4). These distributions established a clinical and modelling context.

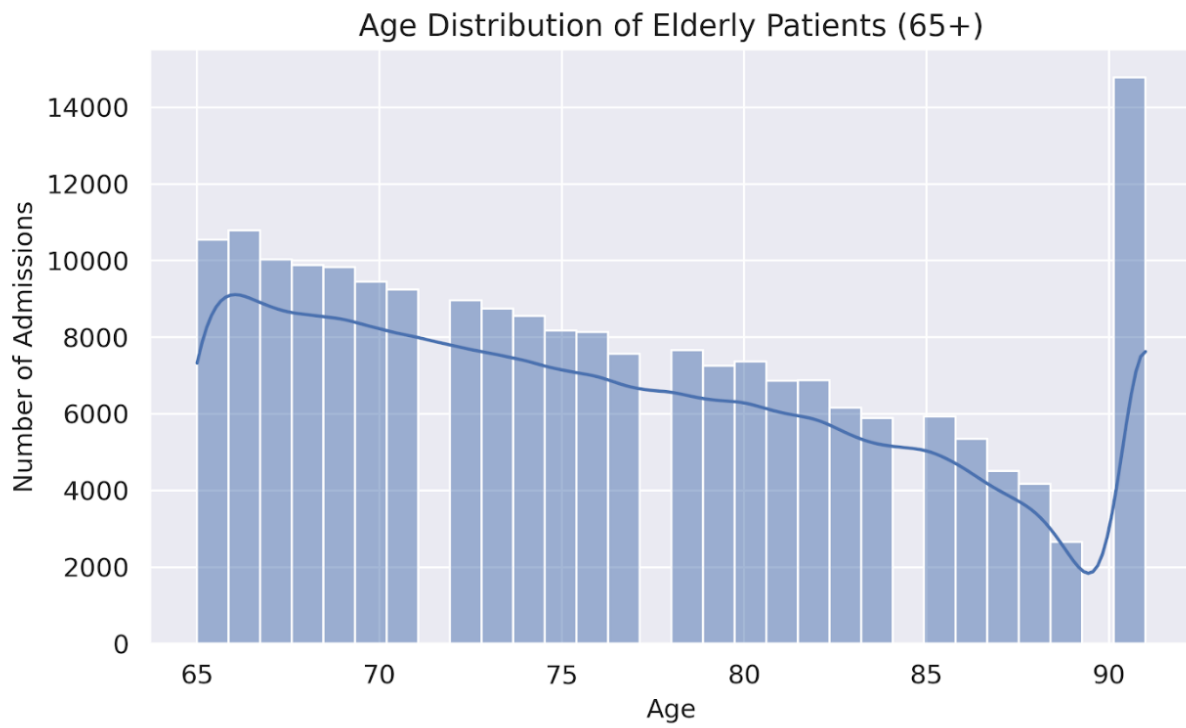


Figure 3. Age distribution of elderly patient admissions (aged 65+) in the study cohort.

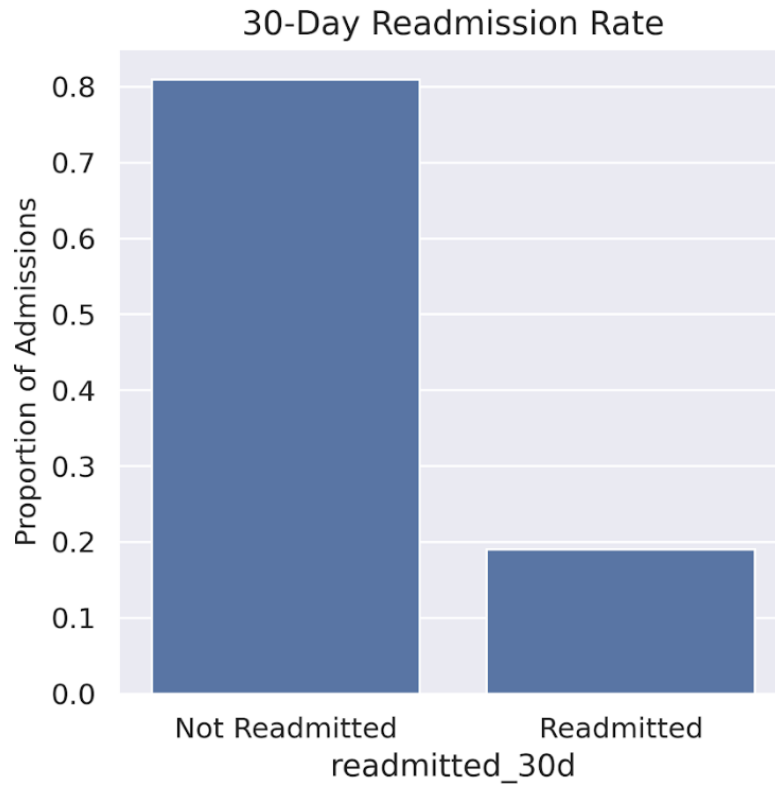


Figure 4. Proportion of admissions resulting in 30-day readmission versus no readmission.

3.3.2 Feature Distributions and Correlations

Correlation analysis helped identify which features were most predictive of readmission and detect redundancy among engineered variables. Temporal features based on prior visits and lab abnormality proportions showed the strongest associations with readmission. As expected, overlapping time-window features (e.g., 30/60/90-day counts) were highly correlated (Figure 5). These insights supported feature selection and modelling decisions.

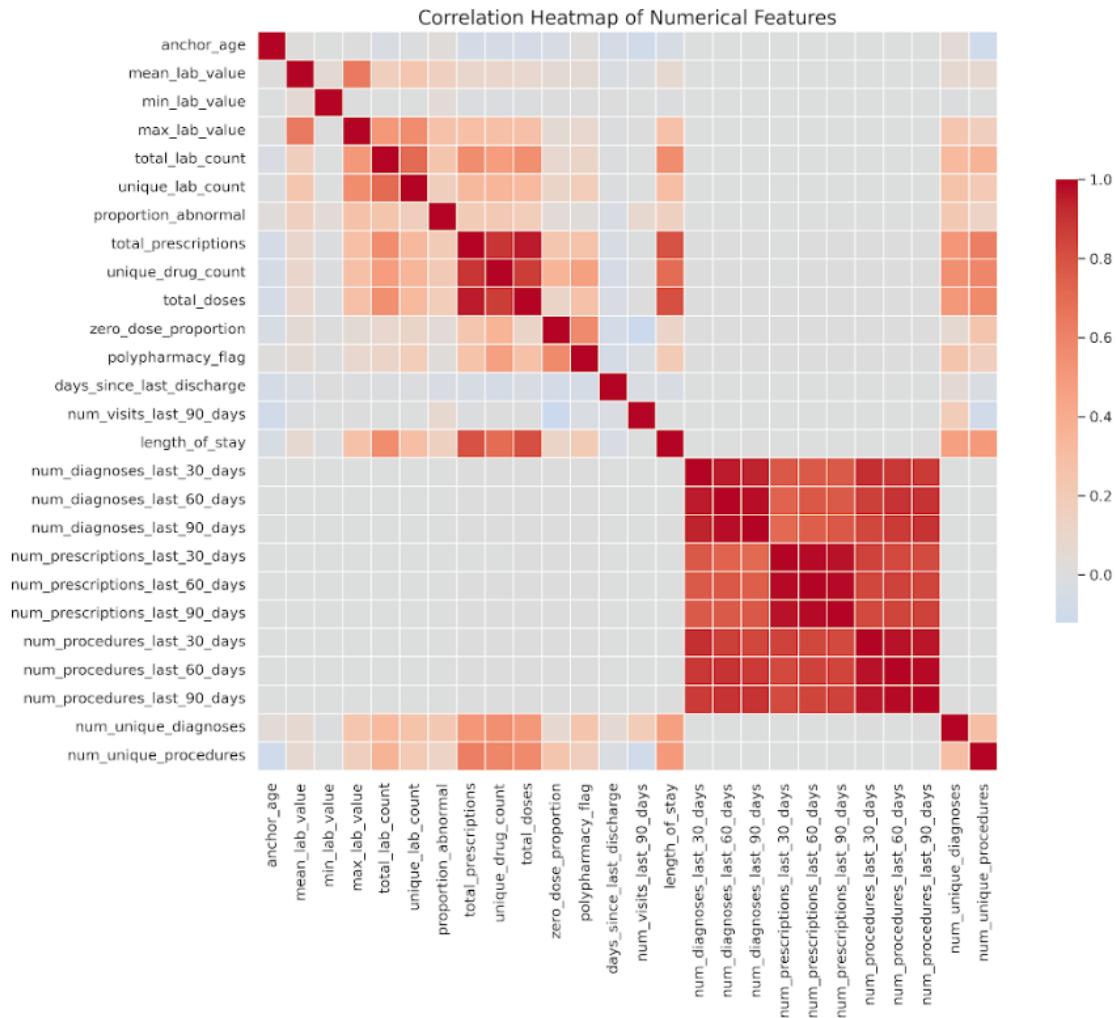


Figure 5. Correlation heatmap of selected numerical features in the structured dataset.

3.3.3 Categorical Predictors and Readmission Risk

Admission and discharge characteristics were important predictors. Patients admitted via emergency or elective routes had higher readmission rates compared to those from outpatient or observation units (Figure 6). Similarly, referral-based, and psychiatric transfers showed elevated risk, reflecting underlying clinical complexity (Figure 7). These variables were retained as categorical inputs in downstream models.

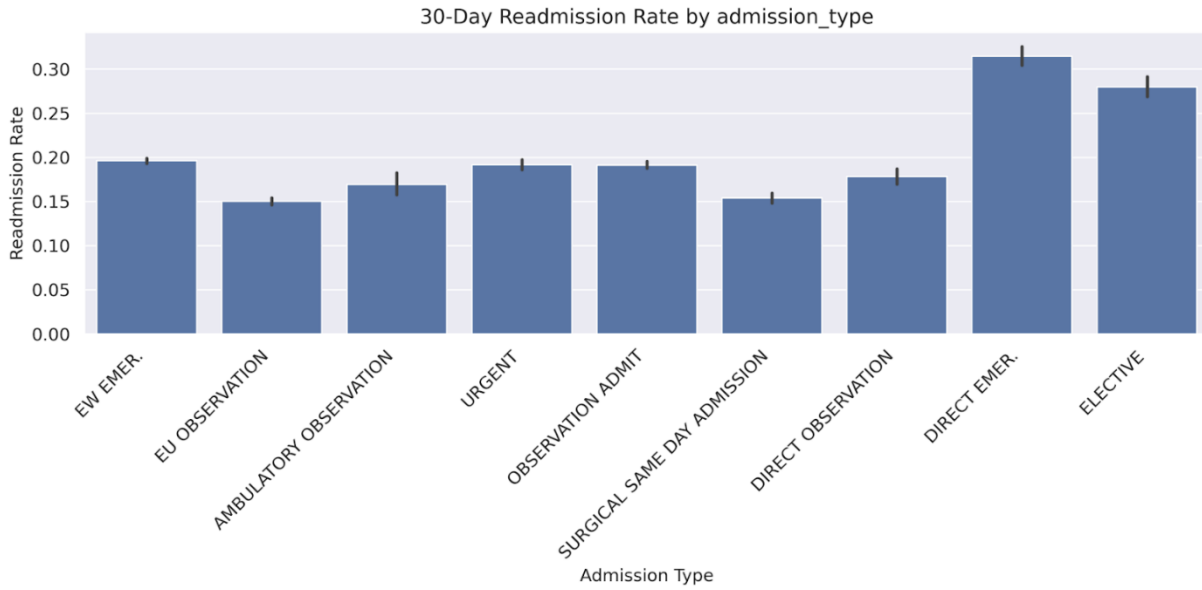


Figure 6. 30-day readmission rate by admission type across the study cohort.

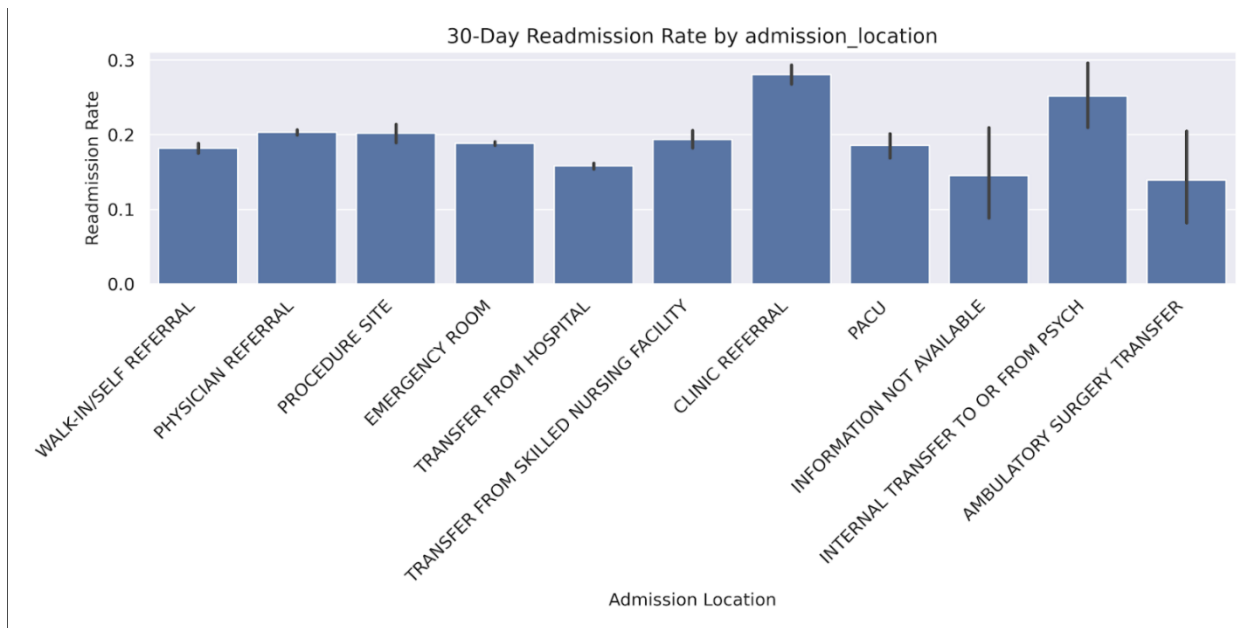


Figure 7. 30-day readmission rate by admission location at time of hospitalization.

3.3.4 Diagnoses and Procedures

Congestive heart failure, hypertension, and atrial fibrillation were common conditions linked to higher readmission risk (Figure 8). Procedures such as central line placement and mechanical ventilation were also associated with elevated risk (Figure 9).

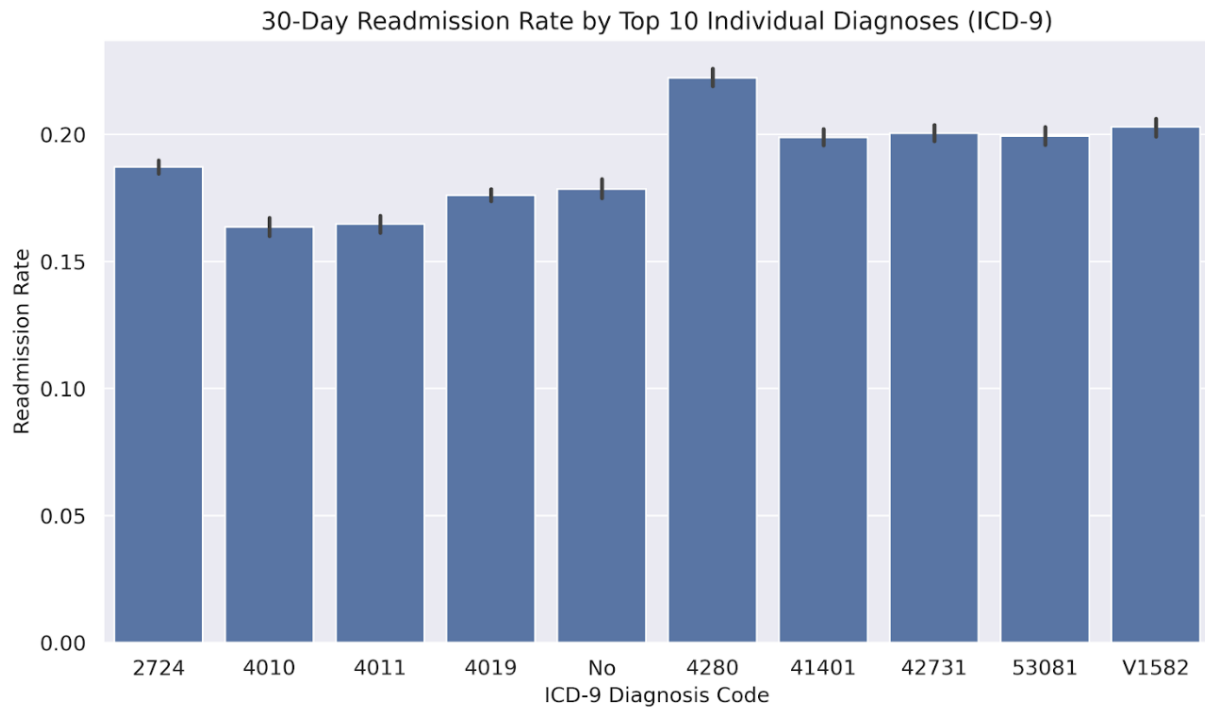


Figure 8. 30-day readmission rate by top 10 most common individual ICD-9 diagnoses.

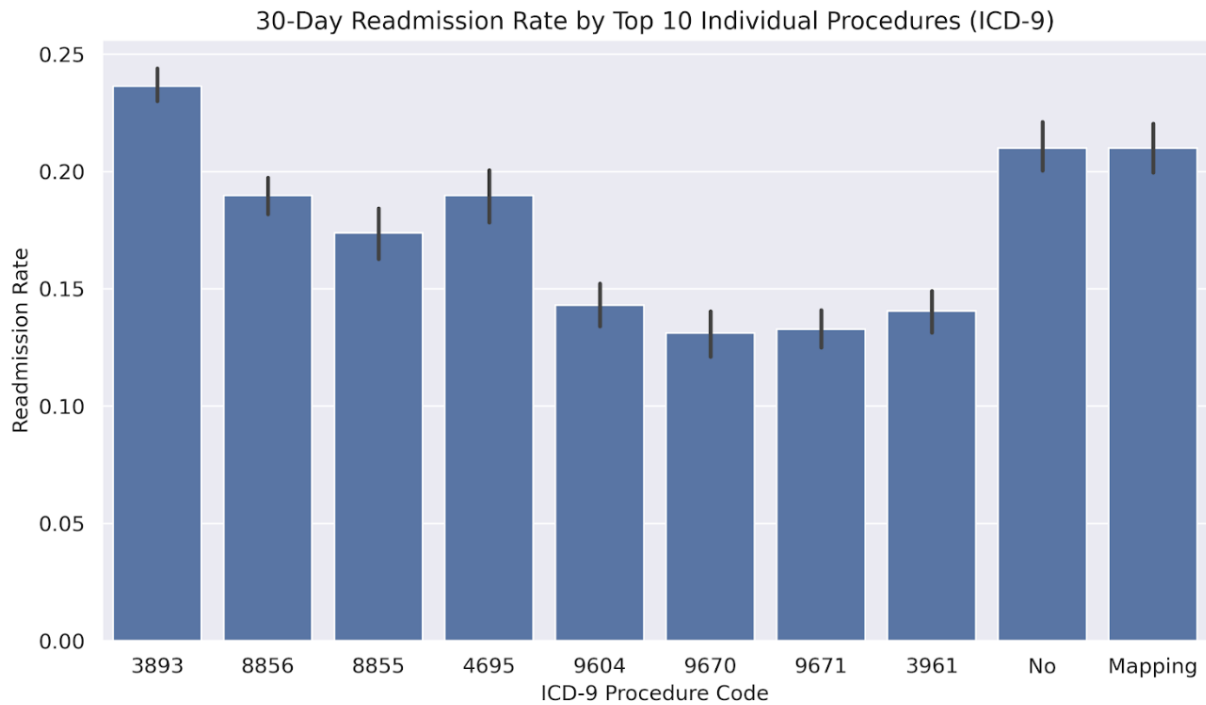


Figure 9. 30-day readmission rate by top 10 most common individual ICD-9 procedures.

3.3.5 Clinical Readmission Patterns

Key clinical signals of readmission risk included polypharmacy, abnormal lab values, and higher medication diversity. These features captured clinical instability and treatment intensity, reinforcing their importance in predictive modelling. The patterns validated the relevance of engineered clinical features used in traditional and deep learning models (Figures 10, 11, 12).

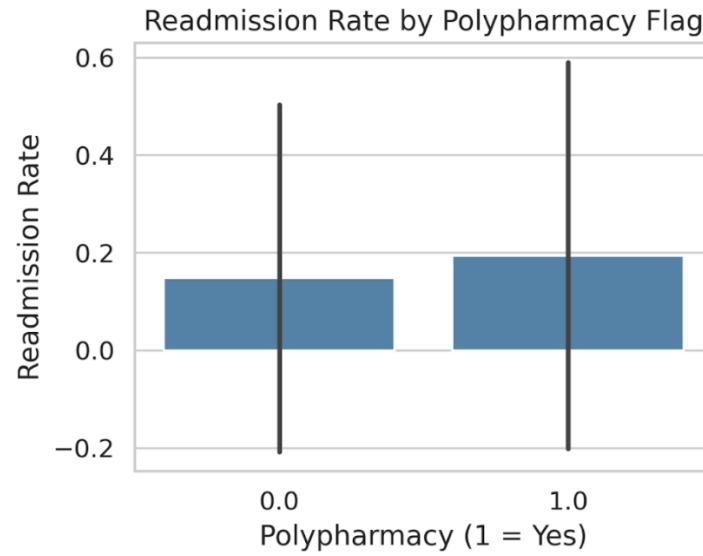


Figure 10. 30-day readmission rate stratified by presence of polypharmacy (≥ 5 concurrent medications).

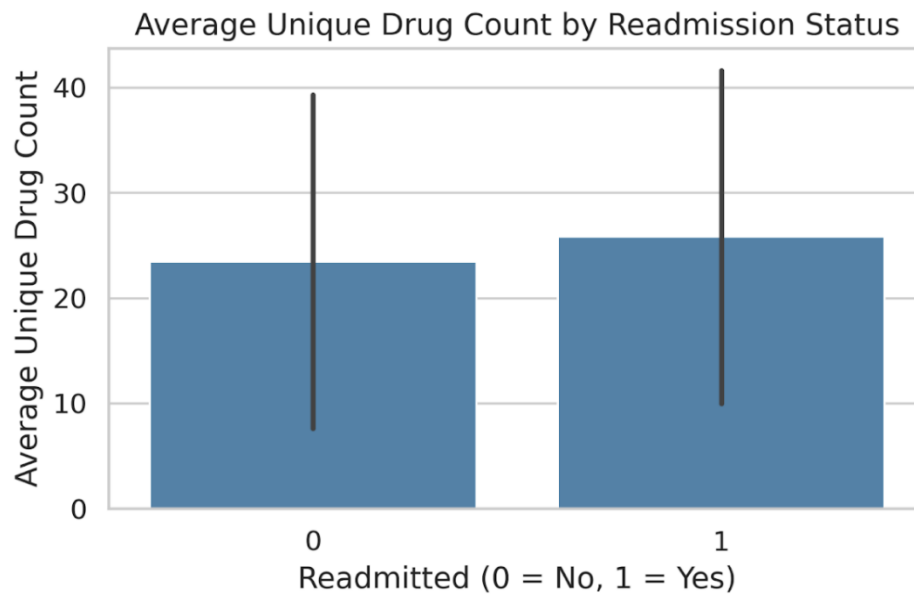


Figure 11. Average unique drug count among readmitted versus non-readmitted patients.

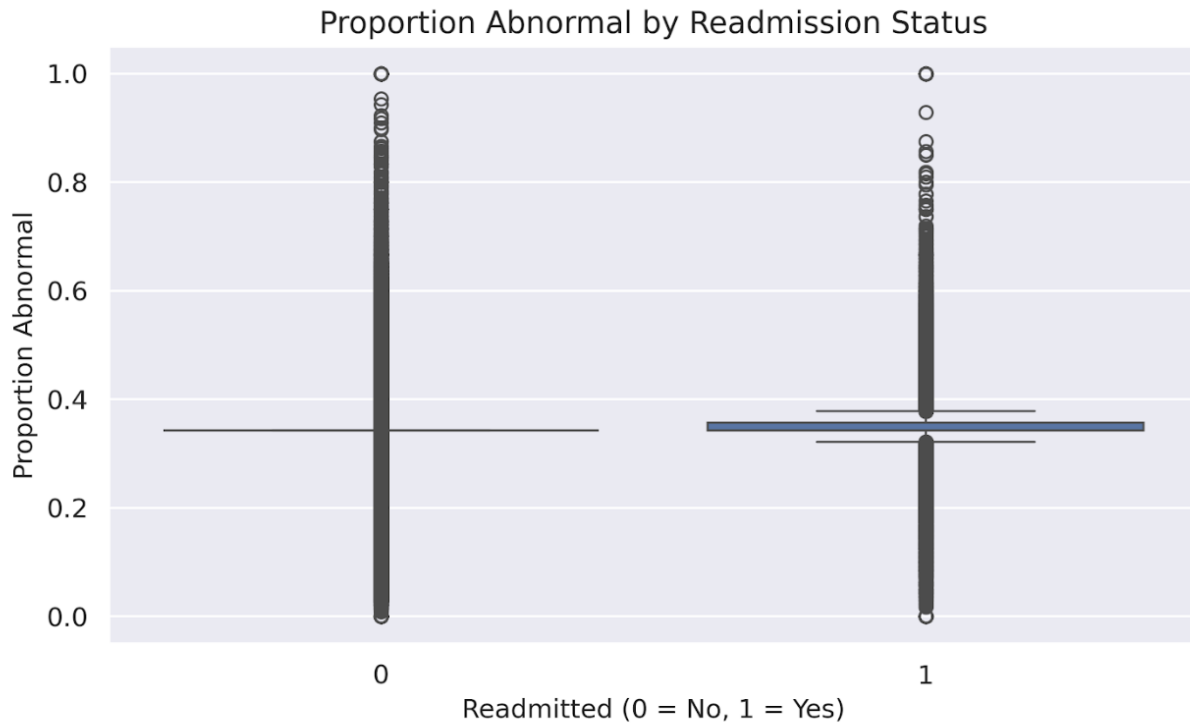


Figure 12. Proportion of abnormal lab results stratified by 30-day readmission status.

3.4 Modelling

This section outlines the predictive modelling strategies to identify 30-day hospital readmissions among elderly patients. The modelling workflow adopts a progressive structure, beginning with baseline machine learning models trained exclusively on structured clinical data. It then advances to deep learning approaches, including a Multi-Layer Perceptron (MLP) and fine-tuned ClinicalBERT applied to discharge summaries. Building on these, hybrid architectures were developed using mid-fusion and late-fusion techniques to integrate structured and unstructured inputs. All models were trained and evaluated using a standardised dataset pipeline to ensure fair and reproducible comparisons. Class imbalance was addressed through SMOTEENN resampling, and performance was assessed using established classification metrics. Finally, explainable AI techniques were introduced to enhance interpretability and support the clinical relevance of model predictions. Hyperparameters for each model were selected through

empirical tuning to balance predictive performance, computational efficiency, and generalisability.

3.4.1 Baseline Models on Structured Data

Logistic Regression (LR) (Cox, 1958), Random Forest (RF) (Breiman, 2001), and XGBoost (XGB) (Chen & Guestrin, 2016) were three baseline models developed on structured EHR data to serve as benchmarks for evaluating the added value of deep learning and hybrid approaches.

Figure 13 outlines the workflow modelling pipeline of the baseline models.

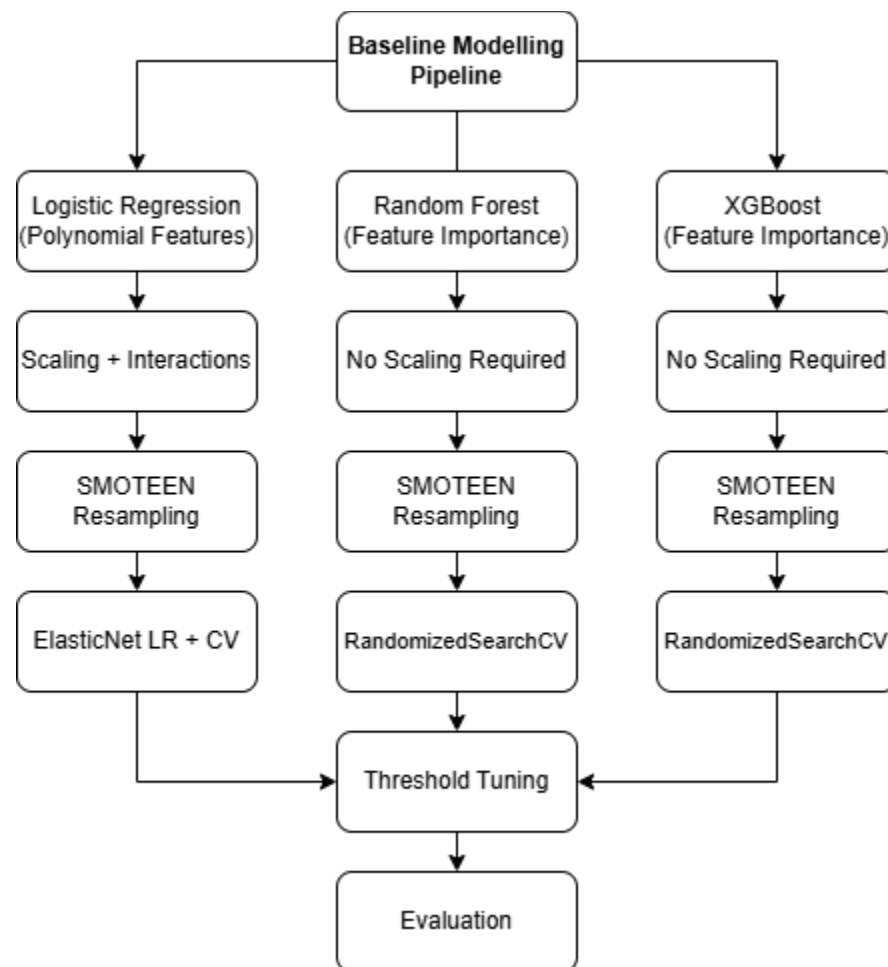


Figure 13. Baseline classical machine learning modelling pipeline for Logistic Regression, Random Forest, and XGBoost models.

3.4.1.1 Data Preparation

Structured features were extracted from the MIMIC-IV database and underwent the following preprocessing steps:

- Variance Thresholding to remove constant and near-constant features.
- Univariate Feature Selection using ANOVA F-score to retain the top 80 predictive variables.
- Missing Value Imputation with median values for numerical fields.
- Standard Scaling using StandardScaler, fitted only on the training set to prevent data leakage.
- Class Balancing with SMOTEENN (sampling strategy = 0.5), combining minority over-sampling and noise cleaning. Batch-wise resampling was used to improve memory efficiency on large datasets.

This pipeline ensured reproducibility and consistency across all baseline models.

3.4.1.2 Logistic Regression (LR)

A regularised logistic regression model was trained using an ElasticNet penalty, balancing L1 and L2 regularisation ($l1_ratio = 0.5$) to promote sparsity while retaining correlated features. PolynomialFeatures (degree = 2) was applied in "interaction-only" mode to enhance the model's ability to capture interactions between features. SMOTEENN was applied only to training data.

Hyperparameters:

- Solver: SAGA (supports ElasticNet and large datasets)
- Penalty: ElasticNet ($l1_ratio = 0.5$)
- Class Weighting: Balanced
- Max Iterations: 200
- Cross-Validation: 3-fold stratified CV on the resampled training data

3.4.1.3 Random Forest (RF)

An ensemble-based Random Forest classifier was optimised using RandomizedSearchCV over a range of parameters, including:

- Number of estimators (trees)
- Maximum tree depth
- Feature sampling strategies (max_features)
- Minimum samples per split

Class imbalance was addressed using class_weight='balanced_subsample', which recalculates weights for each bootstrap sample.

Model Selection:

- Search Method: Randomised search (20 configurations)
- Final Model: Retrained on complete resampled training set using best parameters.

Interpretability was supported via Gini-based feature importance rankings extracted from the trained model.

3.4.1.4 XGBoost (XGB)

The XGBoost classifier was built using XGBClassifier with the histogram-based tree method (tree_method='hist') for faster training on large datasets. Hyperparameter optimization was performed using a randomised search with 3-fold stratified cross-validation.

Class imbalance was mitigated by combining SMOTEENN with scale_pos_weight, set to the inverse ratio of class frequencies in the training set.

Hyperparameters:

- Booster: Gradient boosting trees
- Objective: Binary logistic classification
- Tree Method: hist for speed and memory efficiency

Feature importance scores were saved using gain-based importance to support interpretability and model insights.

3.4.1.5 Evaluation

Baseline models were evaluated on a held-out test set using standard classification metrics: AUROC, F1 Score, Precision, Recall, and the full classification report. To optimise classification under class imbalance, thresholds were tuned using the F1-optimal point from the precision-recall curve rather than a fixed 0.5 cutoff. All predictions, thresholds, and feature importances were saved for benchmarking and use in ensemble models. The same evaluation procedure was applied across all models, including deep learning and fusion approaches, ensuring consistent and fair performance comparisons. These baselines served as interpretable benchmarks for assessing the added value of more complex architectures introduced later.

3.4.2 Deep Learning

To build on the traditional machine learning baselines, this study implemented several deep learning models to capture complex, non-linear relationships in both structured and unstructured clinical data. Neural network architectures were selected for their capacity to model intricate interactions and extract meaningful representations, especially from high-dimensional sources like discharge summaries. This section outlines the design and training of these models, beginning with a Multi-Layer Perceptron (MLP) trained on structured data, followed by ClinicalBERT-based models for text, and concluding with fusion approaches that integrate both data modalities.

3.4.2.1 MLP on Structured Data

To ensure consistency with ClinicalBERT-based models and facilitate fusion architectures, a dedicated Multi-Layer Perceptron (MLP) model (Rumelhart, Hinton, & Williams, 1986) (Figure 15) was trained on structured features filtered to match the ClinicalBERT cohort by `hadm_id` (Figure 14). This ensured all models operated on the same patient subset, enabling fair comparisons across modalities.

Although limited to structured data, this MLP served as a robust baseline to isolate the predictive value of clinical variables. The architecture consisted of three fully connected layers with

decreasing size (512, 256, 128), each followed by Batch Normalization, GELU activation, and Dropout (rates of 0.4, 0.3, 0.2 respectively). This design balanced model capacity with regularization, addressing multicollinearity and overfitting risks common in clinical datasets.

To improve robustness under class imbalance (~19% readmission), the model was trained with Smoothed Focal Loss ($\gamma=2.0$, $\alpha=0.5$, label smoothing=0.1) and class weights computed from the training distribution. This emphasised minority cases while preventing overconfident predictions. Optimization used AdamW with weight decay (0.01), and the learning rate was scheduled using OneCycleLR (max_lr=1e-3) to allow rapid convergence while reducing overfitting. The model was trained with a batch size of 256 and early stopping (patience=5), capped at 30 epochs.

Preprocessing included VarianceThreshold to remove low-variance features, and ANOVA F-score-based SelectKBest to retain the top 80 most predictive variables. Features were standardised using StandardScaler, and SMOTEENN (sampling strategy = 0.4) was applied to the training set to correct imbalance while removing noisy or ambiguous samples. All preprocessing was applied exclusively to the training set to prevent leakage.

Threshold tuning on the validation set was performed using the F1-optimal point on the precision-recall curve, which was then applied to the test set. Probabilities, predictions, and the threshold were saved to support fusion models and reproducible evaluation.

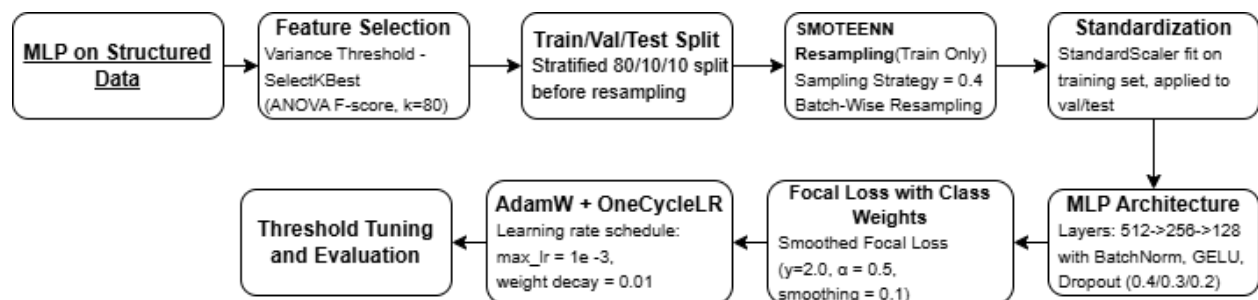


Figure 14. Structured MLP modelling pipeline: feature selection, SMOTEENN resampling, standardization, MLP setup, and evaluation.

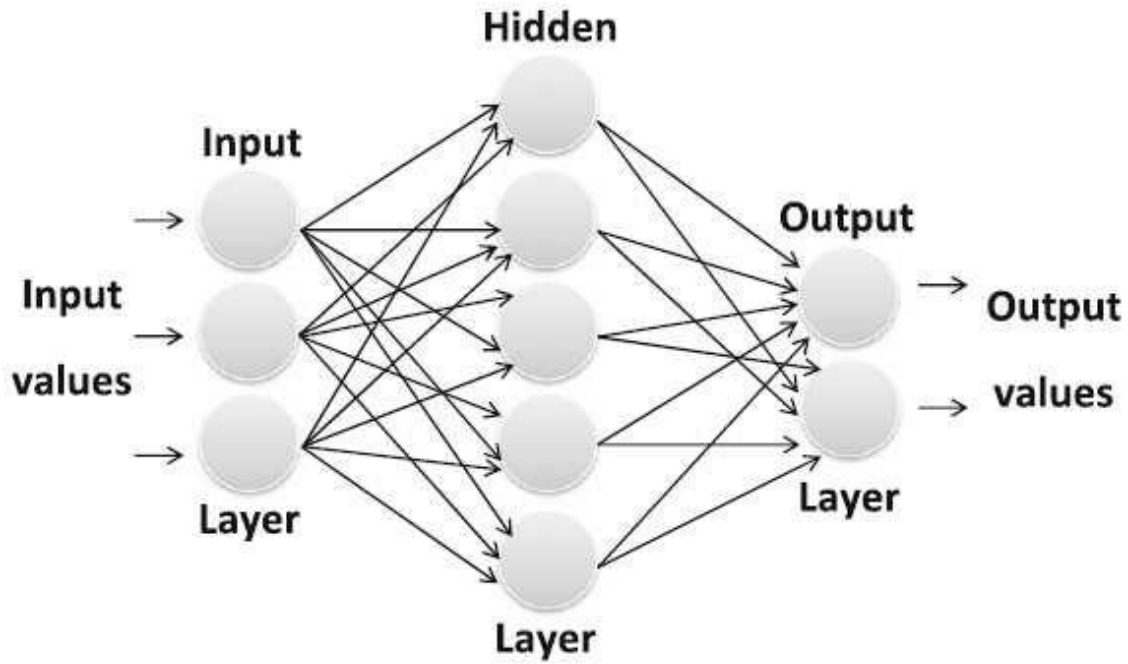


Figure 15. Illustration of a basic Multi-Layer Perceptron (MLP) architecture with input, hidden, and output layers.

3.4.2.2 Fine-Tuned ClinicalBERT on 30-day Readmission

To model unstructured discharge summaries, ClinicalBERT was fine-tuned specifically for 30-day readmission prediction. ClinicalBERT is a domain-adapted variant of BERT, pretrained on MIMIC-III discharge notes to better capture the clinical linguistic style (Alsentzer et al., 2019). In this work, the publicly available Bio_ClinicalBERT checkpoint was employed as the starting point. Discharge summaries were pre-processed in an earlier stage to extract clinically informative sections such as the history of present illness (HPI), hospital course, and discharge instructions using regex-based section filtering. This approach emphasised the most relevant parts of each note for the readmission task.

Given that many MIMIC-IV discharges summaries exceed BERT's maximum input length, a sliding window chunking strategy was adopted. Each document was tokenized and segmented into overlapping chunks of 384 tokens with a stride of 256 tokens, ensuring full coverage without

truncation. Each chunk inherited the parent admission's binary label (readmitted within 30 days vs. not), forming a weakly supervised chunk-level classification setup.

Model Architecture and Training

The Bio_ClinicalBERT model (Figure 17) was initialised, and an attention-based pooling layer was added to aggregate token-level representations into a single embedding per chunk. Unlike original ClinicalBERT fine-tuning, which used only the [CLS] token for classification attention pooling was introduced here to allow the model to learn to prioritise the most informative tokens within each chunk dynamically. A fully connected SoftMax classifier was then used to predict readmission probability from the pooled embedding.

To stabilise training and mitigate overfitting, the first nine transformer layers of ClinicalBERT were frozen, and gradient checkpointing was enabled to optimise GPU memory usage. Consistent with established fine-tuning protocols (Huang et al., 2020), the model was optimised using the AdamW optimiser with an initial learning rate of $2e-5$ and a cosine learning rate schedule with warmup.

To address the class imbalance ($\sim 19\%$ readmitted class), Focal Loss ($\gamma=2.0$) with class weighting was applied instead of standard cross-entropy. This loss function down-weights easy examples and focuses the learning process on harder, minority class instances, improving the model's sensitivity to challenging readmission cases.

Training was conducted with a batch size of 4 and gradient accumulation steps of 2, effectively simulating a batch size of 8 under Colab Pro GPU memory constraints. Mixed-precision (FP16) training was enabled throughout to reduce memory consumption and training time.

Due to Colab runtime limits and the computational intensity of full-length note modelling, the model was fine-tuned for a maximum of three epochs, with early stopping based on validation F1 score applied. However, practical constraints resulted in the model completing two full epochs, with checkpoints saved after each epoch to ensure progress was not lost.

Evaluation and Embedding Extraction

Post-training, chunk-level predicted probabilities were aggregated by averaging across all chunks belonging to the same admission (hadm_id), resulting in admission-level readmission predictions. Performance was evaluated on a held-out test set using metrics including AUROC and PR AUC, following best practices for clinical readmission tasks. Furthermore, to enable downstream multimodal modelling, [CLS] token embeddings were extracted from each chunk, and averaged per admission to produce dense ClinicalBERT representations. These embeddings were subsequently used in mid-fusion and late-fusion architectures combining structured and unstructured patient information. Overall, this fine-tuning strategy enabled full utilization of lengthy clinical notes, prioritised hard-to-classify examples, and generated interpretable, admission-level embeddings critical for downstream predictive modelling (Figure 16).

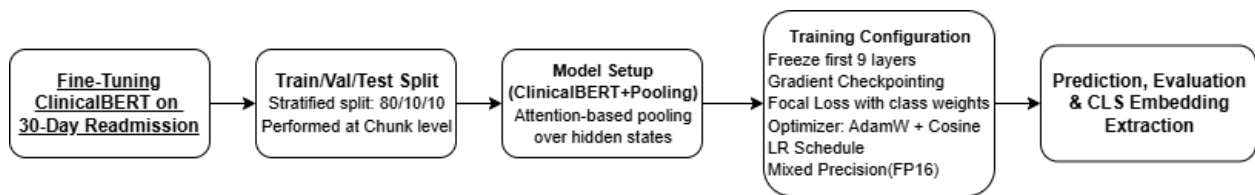


Figure 16. Fine-tuning pipeline for ClinicalBERT on section-filtered discharge summaries for 30-day readmission prediction.

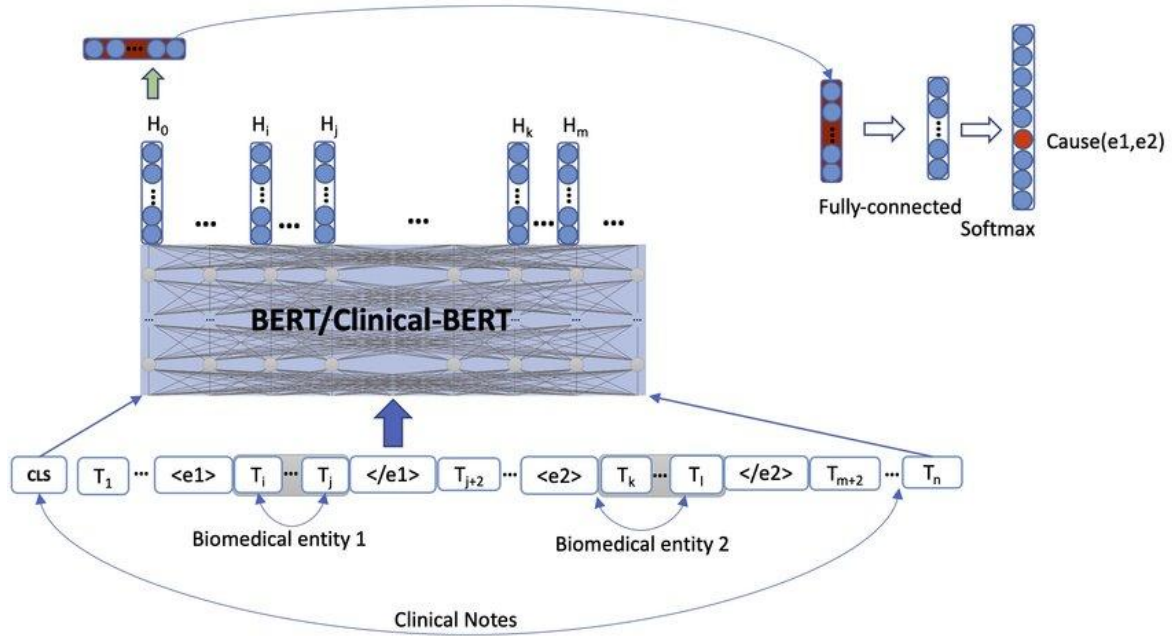


Figure 17. ClinicalBERT architecture for extracting biomedical entity relationships from clinical notes (Khetan et al., 2021).

3.4.2.3 MLP Classifier on ClinicalBERT Embeddings

To isolate the predictive value of unstructured clinical narratives independently from structured variables, a lightweight Multi-Layer Perceptron (MLP) was trained on dense note representations derived from fine-tuned ClinicalBERT embeddings (Figure 18). The embeddings were generated by applying an attention-pooling layer over the final hidden states of ClinicalBERT, which had been fine-tuned for 30-day readmission prediction using Focal Loss and class weighting. This ensured that the input representations were context-aware, clinically optimised, and specifically tailored to the task. Each admission was represented by a single 768-dimensional embedding, obtained by averaging across sliding-window chunks of section-filtered discharge summaries.

Model Architecture and Training

The MLP consisted of three fully connected layers ($768 \rightarrow 512 \rightarrow 256 \rightarrow 2$), interleaved with Batch Normalization, GELU activations, and Dropout (0.3, 0.2). A LayerNorm layer was applied to the input to further stabilise training given the highly contextualised embeddings. The model was trained using Smoothed Focal Loss ($\gamma=2.0$) with class weights computed from the training

distribution (~19% positive cases) to handle class imbalance. Optimization employed AdamW with weight decay (0.01) and a Cosine Annealing learning rate scheduler over 30 epochs. Early stopping with a patience of 7 epochs was used to prevent overfitting. The validation set was used for threshold tuning by selecting the F1-optimal cutoff on the precision-recall curve.

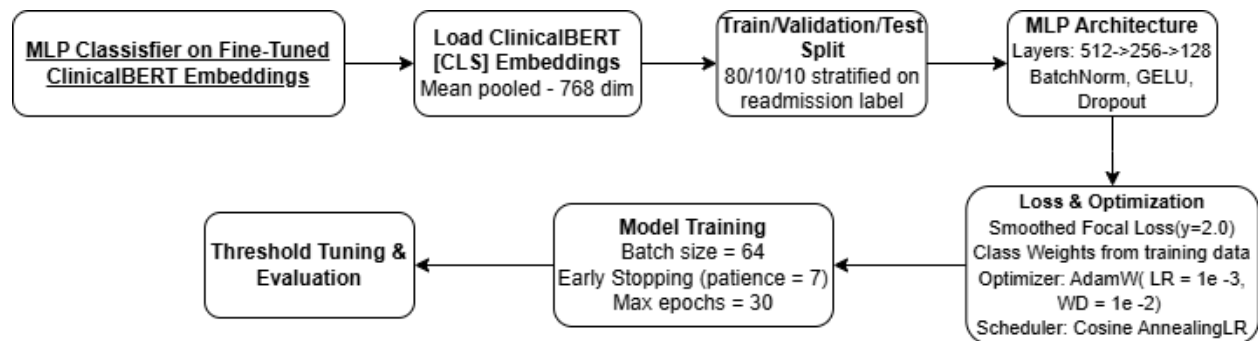


Figure 18. Pipeline for MLP classifier training on mean-pooled [CLS] embeddings from fine-tuned ClinicalBERT.

3.4.2.4 Mid-Fusion Models

To integrate structured and unstructured patient data, a mid-fusion strategy (Baltrušaitis, Ahuja, & Morency, 2019) was adopted using fine-tuned ClinicalBERT embeddings combined with structured EHR features. This hybrid design aims to capture complementary predictive signals from both modalities (Figure 19).

Lightweight MLP-Based Mid-Fusion

The fusion architecture concatenated ClinicalBERT-derived [CLS] token embeddings (averaged across sliding-window chunks) with pre-processed structured features. Each modality was passed through a dedicated MLP branch to project into a shared latent space. An attention gating layer then adaptively weighted the structured and textual representations before final classification.

Architecture and Training Configuration

Each branch used two dense layers with GELU activation, batch normalization, and dropout ($p=0.3$), promoting stable learning and regularization. The attention gating layer ensured dynamic weighting of inputs, balancing interpretability, and performance. The classifier mapped

the fused 128-dimensional representation to a SoftMax output. To address class imbalance and improve generalization, the model was trained with Smoothed Focal Loss ($\gamma=2.0$, label smoothing=0.1). Optimization used AdamW with a learning rate of $1e-3$, weight decay of $1e-2$, and Cosine Annealing LR scheduling over 30 epochs. Early stopping (patience=5) was based on validation F1 score. These choices were informed by prior model iterations and empirical tuning, balancing performance, efficiency, and interpretability.

Alternative Strategy Considered

A deeper, end-to-end mid-fusion model jointly fine-tuning ClinicalBERT with structured features was explored but not implemented due to computational constraints and diminishing returns. Given the effectiveness of the lightweight model and the pre-fine-tuned text encoder, the additional complexity was not justified within the project scope. This approach provided a practical and effective mid-fusion solution, supporting later ensemble strategies and contributing to overall model performance.

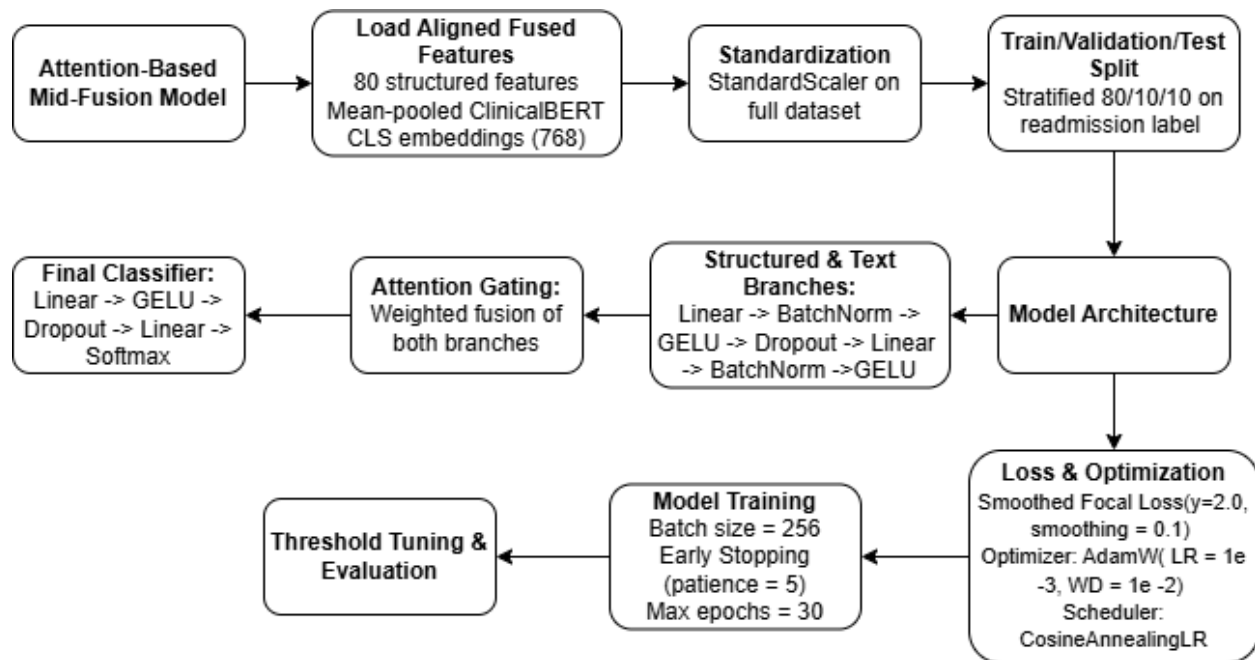


Figure 19. Attention-based mid-fusion modelling pipeline combining structured features and ClinicalBERT embeddings with dynamic modality gating.

3.4.2.5 Late Fusion Model

Predictions were aligned by hospital admission ID (hadm_id) to ensure consistency. Only overlapping admissions in both test sets were retained, and their respective probability scores were concatenated into 2-dimensional feature vectors. These served as input to a meta-classifier, responsible for the final readmission prediction (Figure 20). Late fusion was chosen instead of early fusion because it allows each modality (structured data and text) to be modelled and optimised separately before combining predictions. This approach preserves interpretability (each model's contribution is clear) and flexibility in ensembling, aligning with the study's goal of clinician-friendly AI. Early or mid-fusion strategies were considered, but late fusion offered a simpler integration of modalities given computational constraints and provided better modular interpretability.

A Random Forest classifier was selected as the meta-learner due to its powerful performance with low-dimensional inputs, ability to model nonlinear interactions, and built-in handling of class imbalance via `class_weight='balanced'`. The number of trees was set to 100 to maintain stability while keeping training efficient, and `max_depth=5` was chosen to reduce overfitting on the small input space. These values were selected based on prior validation experiments and follow the best practices for the ensemble of model predictions. Model probabilities were calibrated using the precision-recall curve, and an F1-optimal threshold was selected to balance sensitivity and precision, critical in the readmission prediction context.

This late fusion ensemble outperformed single-modality models by harnessing the independent strengths of both structured and text-based inputs. Unlike early or mid-fusion approaches that combine raw features, late fusion benefits from modular training and improved robustness, as each base model is independently optimised.

Interpretability Considerations

Despite being a black-box ensemble, interpretability was preserved through modular design. Since the final prediction is derived from two distinct sources, structured clinical variables and narrative text, the contribution of each modality remains traceable. This transparency supports trust and decision auditing in clinical settings.

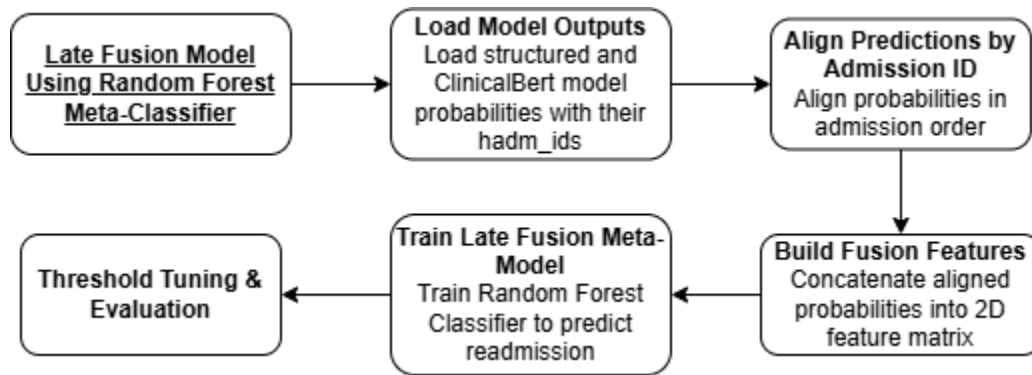


Figure 20. Late fusion modelling pipeline using a Random Forest meta-classifier to combine model outputs.

3.5 XAI and Feature Importance Approaches

Interpretability is crucial for deploying predictive models in healthcare, especially for high-risk populations like elderly patients. To ensure comprehensive insights into model behaviour, both traditional feature importance techniques and modern explainable AI (XAI) methods were applied, tailored to each model architecture.

Baseline Models (Logistic Regression, Random Forest, XGBoost)

For baseline models, feature importance was quantified using model-specific methods. Logistic regression importance was based on the absolute values of standardised coefficients, enabling direct interpretation of associations with 30-day readmission. For Random Forest and XGBoost, importance was assessed via mean decrease in impurity and split gain, respectively. The top 20 influential features were identified in each case, highlighting structured clinical factors predictive of readmission risk.

Fine-Tune ClinicalBERT Model on 30-day Readmission

For the fine-tuned ClinicalBERT model, interpretability was achieved using gradient-based saliency maps. After training, input gradients were computed to estimate token-level importance within discharge summaries. Saliency heatmaps were generated to visualise which words contributed most strongly to the model's output. In addition to qualitative visualisation, token attribution scores were aggregated to identify the top contributing words across selected

examples. This dual approach enabled both local interpretability at the word level and global insight into key semantic patterns learned by ClinicalBERT.

Structured MLP, ClinicalBERT MLP, and Mid-Fusion Models

For the structured MLP, SHapley Additive exPlanations (SHAP) (Lundberg & Lee, 2017) were used with Kernel SHAP to approximate feature attributions, preserving clinical interpretability after feature selection and scaling. The ClinicalBERT MLP model, leveraging attention-pooled [CLS] embeddings from section-filtered summaries, was explained using Integrated Gradients to estimate the contribution of each embedding dimension. While individual dimensions were not directly interpretable, analysing their relative importance revealed insights into the latent clinical representations.

In the mid-fusion model, an internal attention gating mechanism dynamically weighted structured features and textual embeddings. Analysing the distribution of gate values across patients revealed the model's shifting reliance between data modalities depending on available information.

Late-Fusion Model (Meta-Model on Model Outputs)

In the late-fusion setup, a Random Forest meta-classifier was trained on the outputs of the structured MLP and ClinicalBERT MLP. Feature importance in the meta-classifier reflected the relative predictive contribution of each modality, providing model-level transparency.

Summary

By applying a diverse set of explainability methods including standardised coefficients, SHAP, Saliency Heatmaps, Integrated Gradients, attention gating analysis, and meta-model contributions, this study established a multi-level interpretability framework. This was critical for understanding how structured and unstructured EHR data jointly contributed to readmission risk predictions, supporting transparency and clinical trust, which are the key objectives of this study.

Chapter 4 - Findings and Results

This chapter presents the performance outcomes of all models developed in this study. It begins with results from classical machine learning models trained on structured data, followed by deep learning models applied to both structured and unstructured data. The closing section compares hybrid fusion architectures. All models were evaluated on a held-out test set using consistent metrics: AUROC, F1 Score, Precision, Recall, and F1-optimal thresholds.

4.1 Evaluation Metrics

To ensure consistent and fair comparisons across all models, a standard set of evaluation metrics was applied throughout this study. These metrics were selected based on their suitability for binary classification tasks with imbalanced classes, such as 30-day hospital readmission prediction.

- **Area Under the Receiver Operating Characteristic Curve (AUROC):**

AUROC measures the model's ability to distinguish between readmitted and non-readmitted cases across all classification thresholds. A higher AUROC indicates better overall discrimination.

- **F1 Score:**

The F1 score is the harmonic mean of precision and recall, providing a balanced measure that accounts for both false positives and false negatives. It is particularly useful for imbalanced datasets where accuracy may be misleading.

- **Precision:**

Precision quantifies the proportion of positive predictions that were correct, reflecting the model's ability to avoid false positives.

- **Recall:**

Recall measures the proportion of actual readmissions that were correctly identified, highlighting the model's capacity to detect high-risk cases.

- **F1-Optimal Threshold:**

Instead of using a fixed 0.5 decision threshold, each model's predicted probabilities were converted to class labels using the threshold that maximised the F1 score on the

validation set. This approach balances sensitivity and precision and improves interpretability under class imbalance.

All metrics were computed on held-out test sets using the scikit-learn library, and consistent preprocessing pipelines were applied to ensure reproducibility and fair comparison across classical machine learning, deep learning, and hybrid models (Figures 21, 22).

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Figure 21. Formula for computing the F1 score, the harmonic mean of precision and recall.

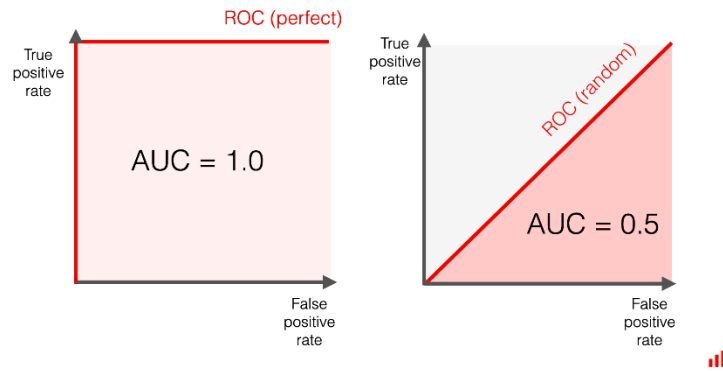


Figure 22. Illustration of perfect and random ROC curves, demonstrating AUC = 1.0 and AUC = 0.5 performance scenarios.

4.2 Baseline Model Results (Structured Data Only)

This section summarises the performance of traditional machine learning models trained solely on structured EHR features. All baseline models underwent consistent preprocessing, feature selection, and SMOTEENN resampling to address class imbalance.

Evaluation was conducted on a held-out test set, with model performance assessed using AUROC, F1 Score, Precision, Recall, and the F1-optimal threshold identified via precision-recall analysis (Figure 4.1- 4.3).

Logistic Regression Results

Logistic Regression was enhanced with ElasticNet regularization and polynomial interaction terms to improve its expressive power. Despite these enhancements, performance remained modest, achieving an AUROC of 0.5973 and an F1 Score of 0.3634. Although recall was high (0.6405), precision was low (0.2536), indicating a tendency to generate false positives.

Random Forest Results

The Random Forest model, tuned via randomised search, demonstrated superior performance compared to Logistic Regression, achieving an AUROC of 0.6655 and an F1 Score of 0.4000. Precision (0.2958) and recall (0.6174) were balanced, highlighting the benefits of ensemble learning for structured clinical data.

XGBoost Results

XGBoost achieved the highest AUROC (0.6677) among baseline models and maintained a competitive F1 Score of 0.3968, with strong recall (0.6579) and moderate precision (0.2840). These results highlight XGBoost's robustness in modelling complex patterns under class imbalance.

Baseline Results Summary

Table 4 provides a comparative overview. Both Random Forest and XGBoost outperformed Logistic Regression, particularly in AUROC and F1 Score, affirming the advantage of non-linear models in this setting. ROC and Precision-Recall curves are shown in Figures 23 and 24.

Table 6. Performance metrics for baseline machine learning models on structured data.

Model	Threshold	Accuracy	Precision	Recall	F1 Score	AUROC	PR AUC
Logistic Regression	0.4968	0.5513	0.2536	0.6405	0.3634	0.5973	0.2572
Random Forest	0.1720	0.6297	0.2958	0.6174	0.4000	0.6655	0.3448
XGBoost	0.0395	0.6001	0.2840	0.6579	0.3968	0.6677	0.3507

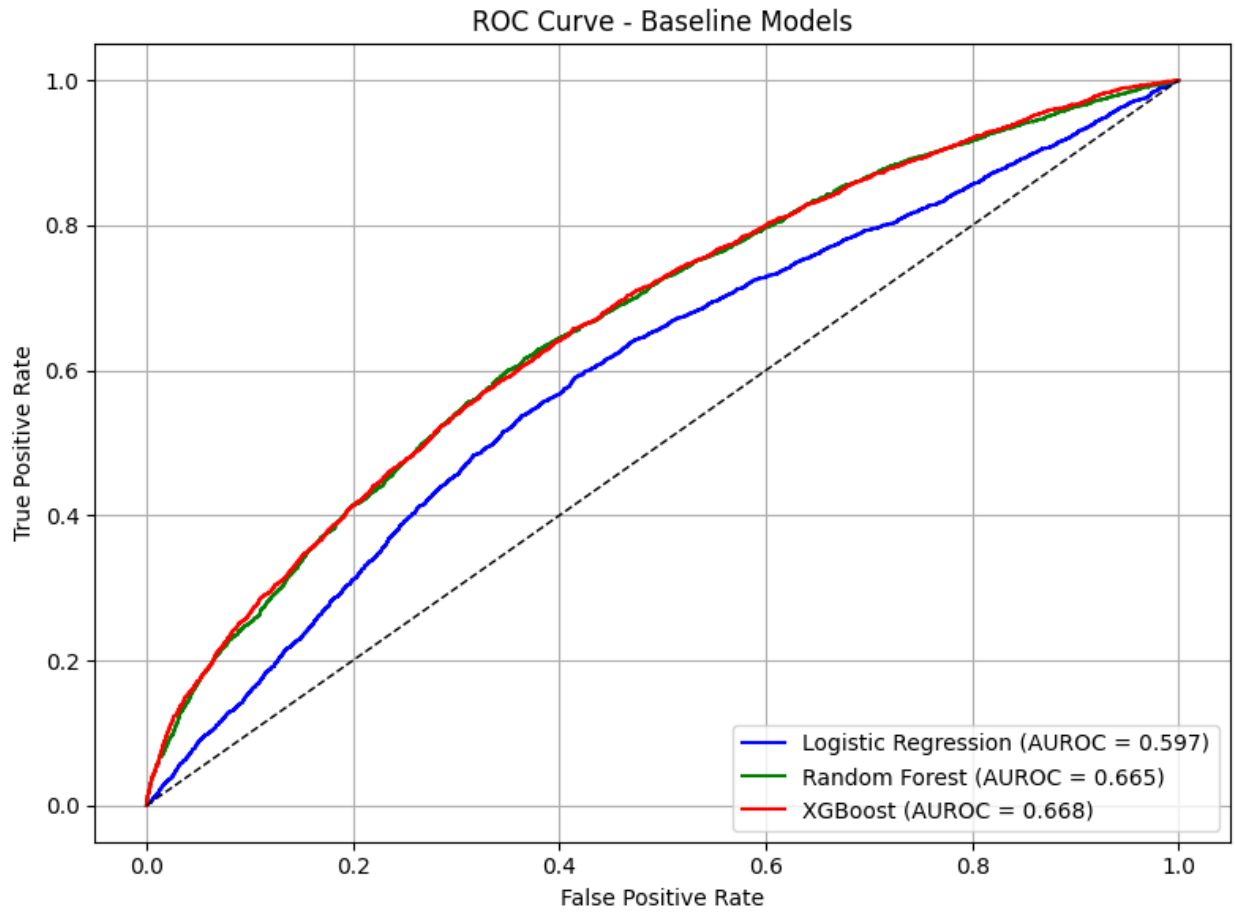


Figure 23. ROC Curve comparing baseline models (Logistic Regression, Random Forest, XGBoost) on structured data.

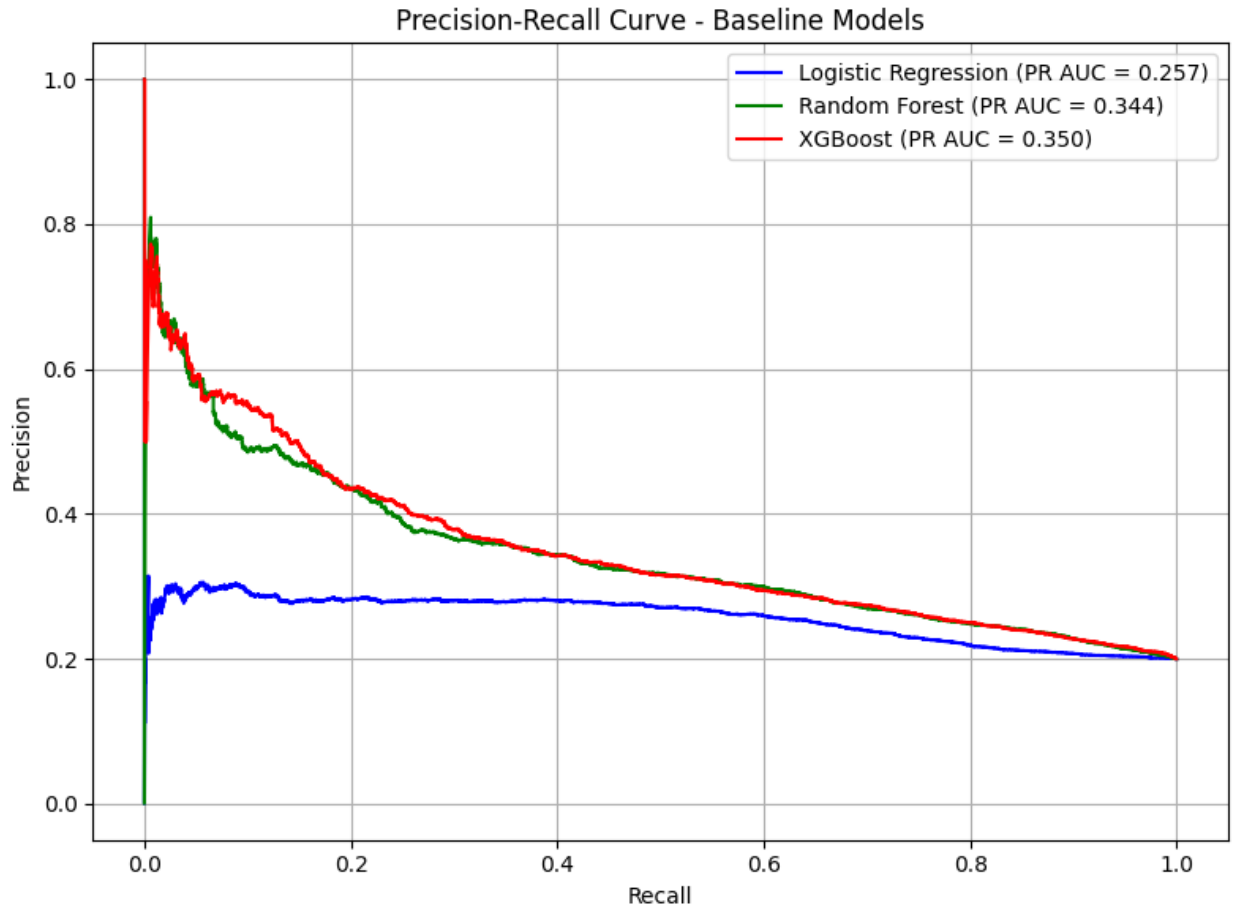


Figure 24. Precision-Recall Curve comparing baseline models (Logistic Regression, Random Forest, XGBoost) on structured data.

4.3 Deep Learning Models Results

This section presents the results of deep learning models trained on structured and unstructured data. All models were evaluated using consistent test sets, with classification thresholds tuned for F1-optimal performance.

Structured MLP Results

The structured MLP model, trained on selected features, achieved an AUROC of 0.6656 and an F1 Score of 0.3963. Recall was 0.6263 and precision was 0.2898 at the optimal threshold of

0.4866, comparable to tree-based baselines but showing a slight edge in recall-focused performance.

Fine-Tuned ClinicalBERT on 30-Day Readmission Results

ClinicalBERT was fine-tuned directly on section-filtered discharge summaries using sliding window chunking and attention pooling. The transformer model achieved an AUROC of 0.6539 and an F1 Score of 0.3899, with recall of 0.5121 and precision of 0.3149. These results demonstrate the transformer's strong ability to extract predictive information from unstructured narratives (Figures 25, 26).

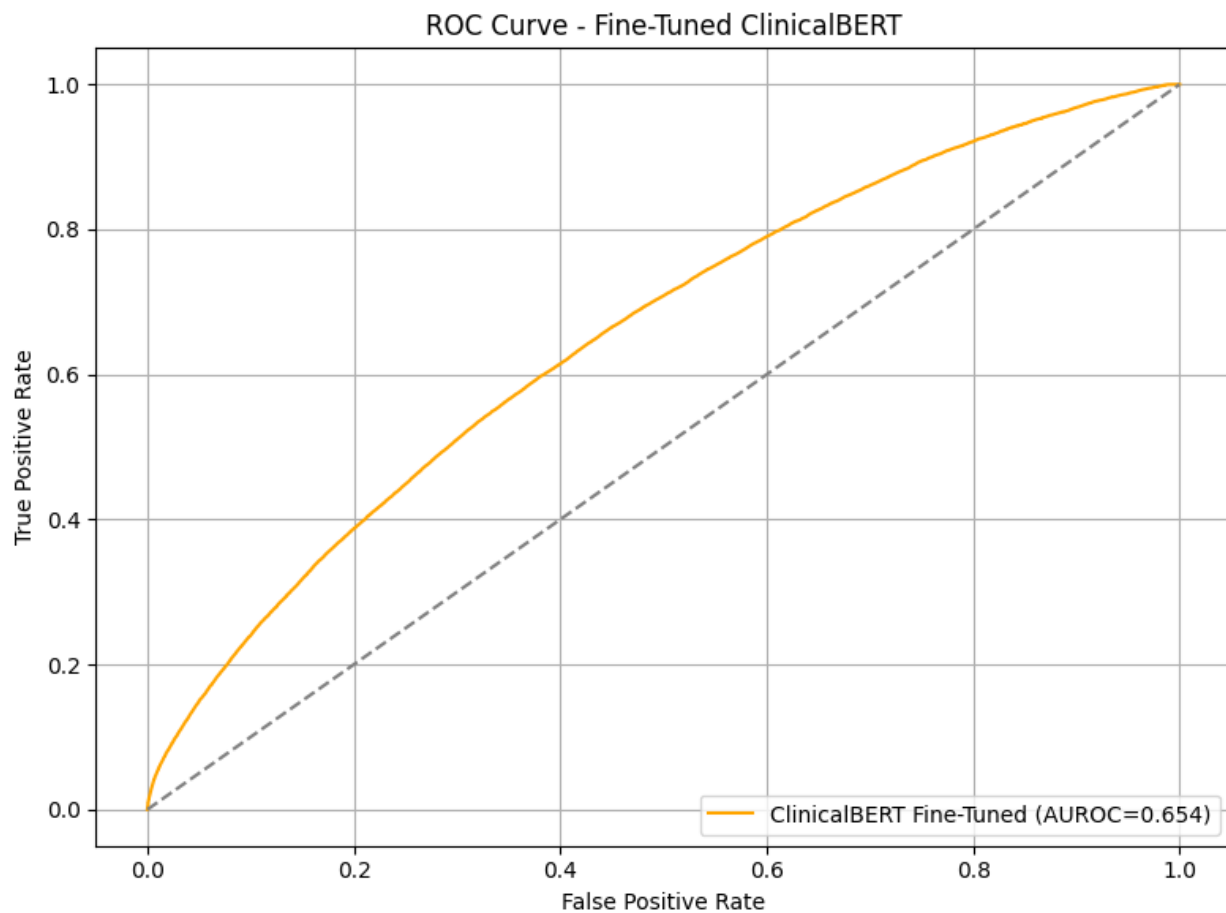


Figure 25. ROC Curve for fine-tuned ClinicalBERT model on predicting 30-day readmissions.

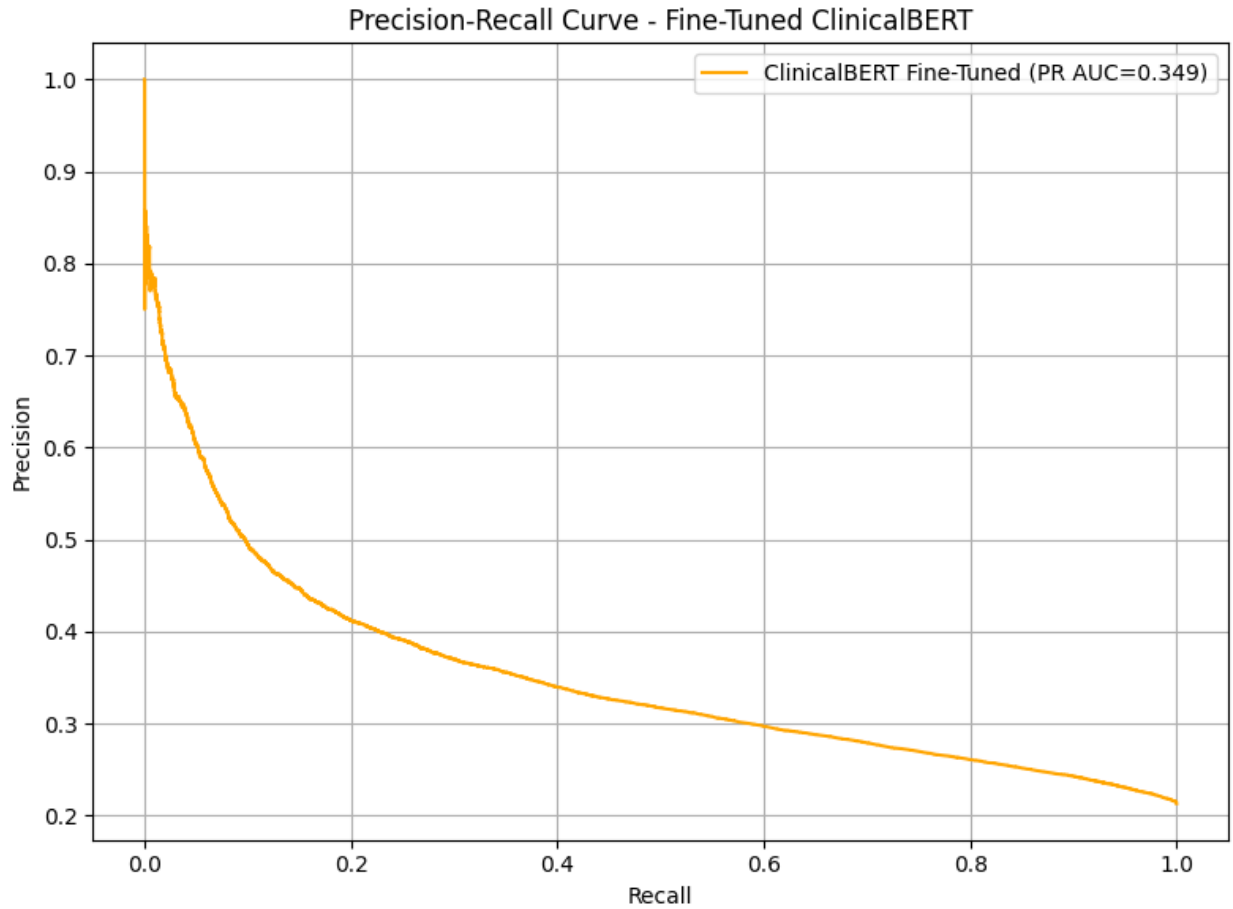


Figure 26. Precision-Recall Curve for fine-tuned ClinicalBERT model on predicting 30-day readmissions.

MLP on Fine-Tuned ClinicalBERT Attention-Pooled Embeddings Results

After fine-tuning ClinicalBERT, mean attention-pooled [CLS] embeddings were extracted per admission. A lightweight MLP classifier was then trained on these frozen representations. The MLP on ClinicalBERT embeddings achieved an AUROC of 0.7418 and an F1 Score of 0.4568, showing improved performance over the full transformer, but with significantly reduced computational overhead during inference.

Deep Learning Results Summary

The deep learning approaches, MLP on structured data, transformer fine-tuning, and embedding-level MLP, can outperform baseline models. These results illustrate that both modalities individually provide strong predictive signals for elderly readmissions.

4.4 Fusion Models Results

This section presents results from hybrid fusion models that combine structured EHR features with unstructured discharge summary embeddings. These models aim to leverage the complementary strengths of both data modalities to enhance predictive performance beyond what either can achieve alone. Fusion was implemented using two distinct strategies: mid-fusion via embedding concatenation and attention gating, and late-fusion via ensemble-level decision modelling. All fusion models were evaluated on the same held-out test set used in earlier sections, and classification thresholds were tuned on the validation set using the F1-optimal point on the precision-recall curve.

Mid-Fusion Results

The mid-fusion model combined structured features and ClinicalBERT embeddings via resolute MLP branches and a dynamic attention gating mechanism. It achieved an AUROC of 0.7517, an F1 Score of 0.4656, recall of 0.6409, and precision of 0.3656 at the optimal threshold of 0.4055. Performance improvements over single-modality models demonstrate the synergistic value of combining structured and unstructured data at the representation level.

Late Fusion Results

The late-fusion approach trained a Random Forest meta-classifier on the probability outputs of the structured and ClinicalBERT MLPs. This ensemble achieved the strongest overall performance, with an AUROC of 0.7608, an F1 Score of 0.4774, precision of 0.4042, and recall of 0.5829, using an optimal threshold of 0.5710. This improvement emphasises the robustness of ensemble-level decision fusion when combining heterogeneous modalities.

Fusion Summary

Both mid-fusion and late-fusion models outperformed all single-modality models. Notably, late-fusion achieved the highest AUROC and PR AUC, further supporting the hypothesis that clinical narratives provide complementary information to structured features. ROC and Precision-Recall curves for all models are presented in Figures 27 and 28, and a comparison table of deep learning and fusion models are displayed in Table 5 below.

Table 7. Performance comparison of Structured MLP, ClinicalBERT MLP, Late Fusion, and Mid-Fusion models.

Model	Threshold	Accuracy	Precision	Recall	F1 Score	AUROC	PR AUC
Structured MLP	0.4866	0.6185	0.2898	0.6263	0.3963	0.6656	0.3407
ClinicalBERT MLP	0.4942	0.6966	0.3557	0.6384	0.4568	0.7418	0.4350
Late Fusion (RF)	0.5710	0.7449	0.4042	0.5829	0.4774	0.7608	0.4661
Hybrid Fusion (MLP)	0.4055	0.7059	0.3656	0.6409	0.4656	0.7517	0.4457

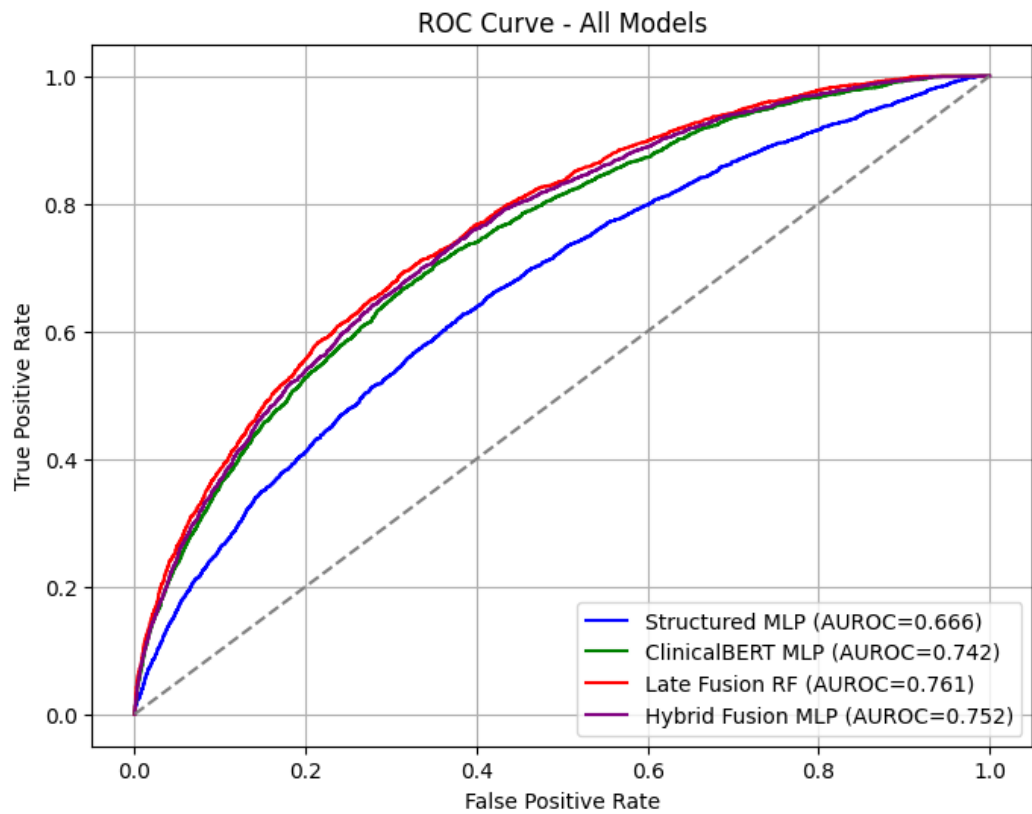


Figure 27. ROC Curve comparing all models (Structured MLP, ClinicalBERT MLP, Late Fusion RF, Hybrid Fusion MLP).

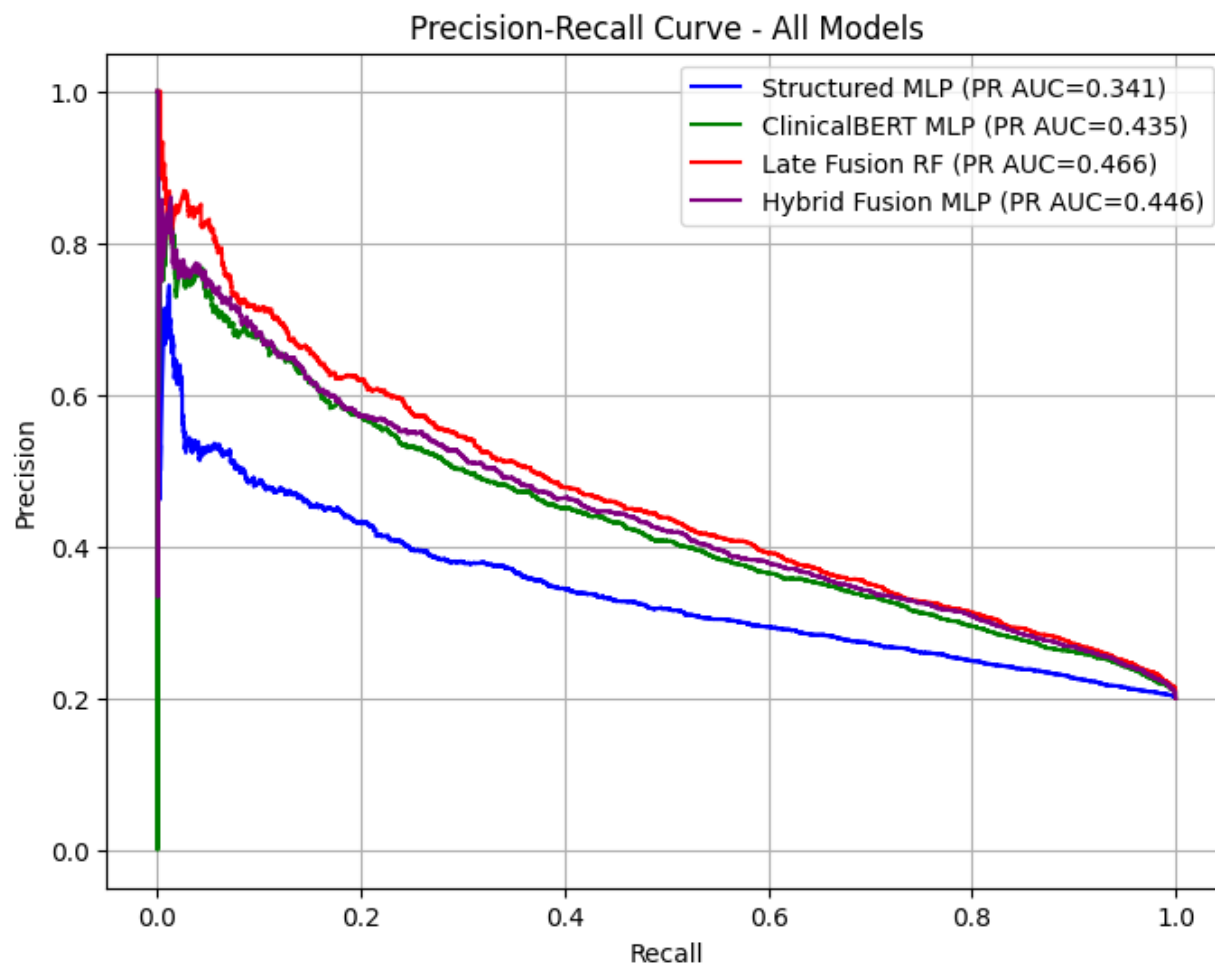


Figure 28. Precision-Recall Curve comparing all models (Structured MLP, ClinicalBERT MLP, Late Fusion RF, Hybrid Fusion MLP).

4.5 Explainability and Feature Importance Results

Explainability analyses were conducted across all models to better understand feature contributions and modality reliance. In the baseline models, key structured predictors consistently emerged across different algorithms. Features such as `days_since_last_discharge`, `num_visits_last_90_days`, and `discharge_location_SKILLED NURSING FACILITY` were identified as highly influential. In the Logistic Regression model, interaction terms involving healthcare utilization and medication burden (e.g., `total_prescriptions x num_visits_last_90_days`) were also significant.

For the structured MLP model, SHAP analysis reaffirmed the prominence of these structured variables, highlighting the continued predictive strength of post-discharge healthcare engagement and medication burden in readmission risk (Figure 29).

Interpretation of the ClinicalBERT MLP using Integrated Gradients revealed that a small subset of embedding dimensions contributed disproportionately to the model's outputs. While individual embeddings are not directly human-interpretable, the findings indicate the model effectively captured important semantic patterns from discharge summaries (Figure 30).

Saliency heatmap visualizations highlighted that ClinicalBERT assigned higher attention to clinically meaningful terms, particularly those related to acute deterioration ("glucose," "seizure," "discharged") and social circumstances ("family") (Figures 31, 32, 33). The corresponding table of top words and their importance scores (Table 6) reinforced these findings, with "patient," "glucose," and "labs" consistently ranked among the most influential. Together, the heatmaps and word importance in relation to 30-day readmission demonstrated that ClinicalBERT effectively prioritised key medical and social factors relevant to 30-day readmission risk.

In the mid-fusion model, gate attention analysis showed that the model dynamically balanced structured features and text embeddings, slightly favouring text embeddings data overall (Figure 34). However, substantial variability was observed across patients, demonstrating the value of flexible multi-modal learning in elderly populations where data completeness can vary.

Finally, the late-fusion Random Forest meta-model showed that both base models contributed meaningfully to the final prediction, with ClinicalBERT MLP probabilities carrying more importance than structured MLP probabilities (Figure 35). This further supports the complementary role of clinical narratives in augmenting structured risk prediction.

Overall, the explainability results confirm that both structured EHR fields and unstructured clinical notes provide important, complementary information for predicting 30-day hospital readmissions among elderly patients.

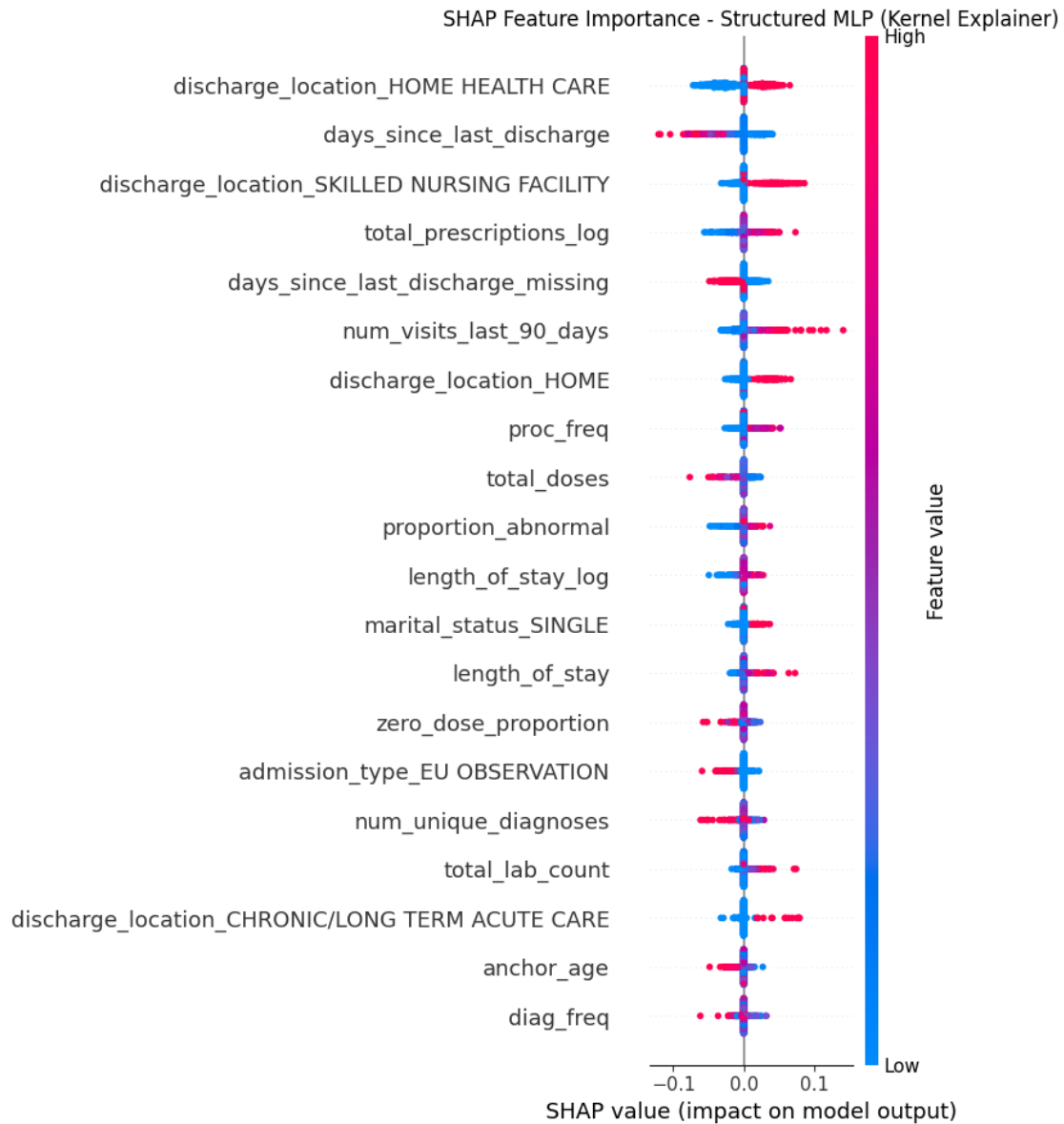


Figure 29. SHAP Feature Importance for Structured MLP.

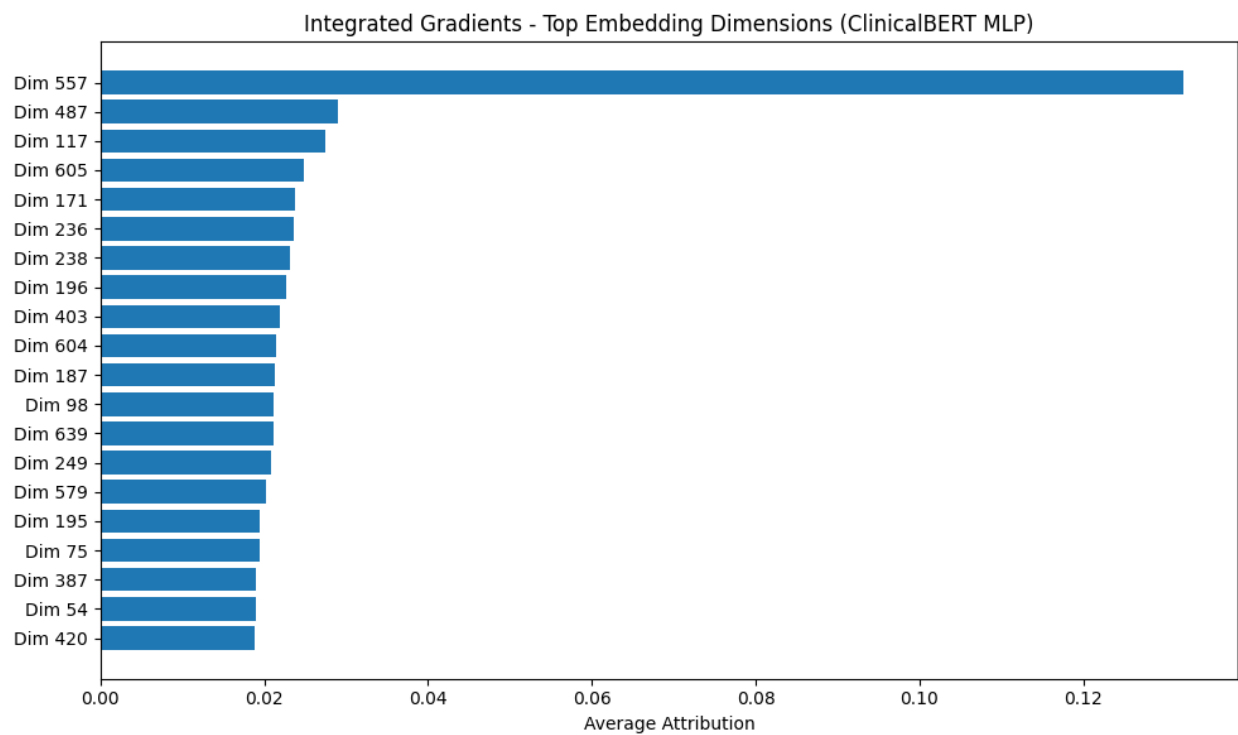


Figure 30. Integrated Gradients - Top Embedding Dimensions (ClinicalBERT MLP).

\ Example 1
 history of present illness : with pm h significant for m ca in far ct in with r hem
 ip leg ia and a pha sia . ydro ce pha lus s p v p s hun
 and multiple u that was transferred from in the setting of altered mental status and concern for
 seizure . patient lives at nursing home facility and requires ass ita nce with all ad ls including feeding at
 base line . the patient is also un res po ns ive to any commands or voice and cannot express
 herself verbal ly . per her h c p . patient was being fed on and noted to be increasingly
 so m no lent . she was allowed to go to sleep . the morning of she was
 found to be entirely un res po ns ive with a te mp to 103 . she was given a
 pa p and transferred at 99 . 4 . 103 . 69 . and
 on room air . she was noted to be o bt and and uro exam significant for s
 lug gis hly reactive pupils . horizontal n ys tag mus . and media left eye de viation . c
 b c w n . la ct ate 4 . 4 which improved to 2 . 6 on re
 che ck . glucose 290 and co 3 20 . ch em - 10 otherwise w n .
 c x r un rem ark able . showed no acute processes . given
 g c ef tri ax one and g van co for u in setting of re current u
 the past . remarkable for 3 bacteria . also given i m g at iva n
 . ne uro consulted in ed . and of concern for re current seizure s . patient was transferred
 ed . of note d n r d ni status was re s cin ded due to family disputes
 in the ed . initia vs 94 139 56 18 98 . a g es of 3 on
 arrival .
 history of present illness : with pm h significant for m ca in far ct in with r hem
 ip leg ia and a pha sia . ydro ce pha lus s p v p s hun
 and multiple u that was transferred from in the setting of altered mental status and concern for
 seizure . patient lives at nursing home facility and requires ass ita nce with all ad ls including feeding at
 base line . the patient is also un res po ns ive to any commands or voice and cannot express
 herself verbal ly . per her h c p . patient was being fed on and noted to be increasingly
 so m no lent . she was allowed to go to sleep . the morning of she was
 found to be entirely un res po ns ive with a te mp to 103 . she was given a
 pa p and transferred at 99 . 4 . 103 . 69 . and
 on room air . she was noted to be o bt and and uro exam significant for s
 lug gis hly reactive pupils . horizontal n ys tag mus . and media left eye de viation . c
 b c w n . la ct ate 4 . 4 which improved to 2 . 6 on re
 che ck . glucose 290 and co 3 20 . ch em - 10 otherwise w n .
 c x r un rem ark able . showed no acute processes . given
 g c ef tri ax one and g van co for u in setting of re current u
 the past . remarkable for 3 bacteria . also given i m g at iva n
 . ne uro consulted in ed . and of concern for re current seizure s . patient was transferred
 ed . of note d n r d ni status was re s cin ded due to family disputes
 in the ed . initia vs 94 139 56 18 98 . a g es of 3 on
 arrival .

Figure 31. Saliency Visualization of Notes - Example 1.

\ Example 2

Figure 32. Saliency Visualization of Notes - Example 2.

of present illness | is a year old man with a pm h of q uad rip leg
 status post - m v c years ago , seizure disorder , ne uro gen bladder s r su
 pra pu bic cat he ter and multi dr ge resistant | presenting as a transfer from
 for seizure | and h y pot ens ion , patient was noted to have 2 grand
 ma | seizure today , for which he was brought to | his initial b p was | there
 he was loaded with k ep pra , and started on van c and z osy | for | ?
 u | he got 3 | v f . patient was noted to be h y pot ens
 ive , for which a c v | was placed and he was started on nor ep ine ph rine
 , la cate was | . 4 . he was transferred to for ne uro gen evaluation | on arrival
 | patient was noted to be so m no lent with an end - tida between 10 and 25 |
 poorly re sp ons ive , he was subsequently | tub ated for air way protection | patient had recent
 admission for abdominal pain , found to have mild la ctic acid osis , treated for | although urine
 cultures contaminated | discharged on | was treated with z osy | but no a b x continued on discharge
 , also noted to have intermittent h y pot ens ion that resolved without intervention | ed course notable for
 | vs : | 97 | 0 h r 70 b p 124 67 r | 18 s po
 | 97 | exam | uad rip leg | male , so m no lent ben ign card
 io pu lm ona ry exam | ign abdominal exam | no fatigue d to answer orientation questions , labs :
 | w b c 9 | 8 - h g 10 | 2 - e r 0 | 5 -
 mg 1 | 6 - e k 45 - la ct ate 1 | 9 - |
 g le

of present illness | is a year old man with a pm h of q uad rip leg
 status post - m v c years ago , seizure disorder , ne uro gen bladder s r su
 pra pu bic cat he ter and multi dr ge resistant | presenting as a transfer from
 for seizure | and h y pot ens ion , patient was noted to have 2 grand
 ma | seizure today , for which he was brought to | his initial b p was | there
 he was loaded with k ep pra , and started on van c and z osy | for | ?
 u | he got 3 | v f . patient was noted to be h y pot ens
 ive , for which a c v | was placed and he was started on nor ep ine ph rine
 , la cate was | . 4 . he was transferred to for ne uro gen evaluation | on arrival
 | patient was noted to be so m no lent with an end - tida between 10 and 25 |
 poorly re sp ons ive , he was subsequently | tub ated for air way protection | patient had recent
 admission for abdominal pain , found to have mild la ctic acid osis , treated for | although urine
 cultures contaminated | discharged on | was treated with z osy | but no a b x continued on discharge
 , also noted to have intermittent h y pot ens ion that resolved without intervention | ed course notable for
 | vs : | 97 | 0 h r 70 b p 124 67 r | 18 s po
 | 97 | exam | uad rip leg | male , so m no lent ben ign card
 io pu lm ona ry exam | ign abdominal exam | no fatigue d to answer orientation questions , labs :
 | w b c 9 | 8 - h g 10 | 2 - e r 0 | 5 -
 mg 1 | 6 - e k 45 - la ct ate 1 | 9 - |
 g le

Figure 33. Saliency Visualization of Notes - Example 3.

Table 8. Important Words per Example (Word Importance Table).

Example	Word	Importance Score
Example 1	patient	3.10598
	glucose	3.10598
	tis	1.84596
	global	1.28371
	multiple	1.28371
	concern	1.28371
	requires	1.28371
	cannot	1.28371
	transferred	1.28371
	medial	1.28371
Example 2	cop	6.25170
	family	6.25170
	physical	4.59288
	social	4.20822
	grave	4.08454
	disease	4.08454
	decreased	4.08454
	mildly	4.08454
	illness	3.60953
	developed	3.56014
Example 3	patient	10.36227
	discharged	10.36227
	labs	10.36227
	disorder	4.46323
	genic	4.43423
	logical	4.43423
	ben	4.05481
	too	4.05481
	ated	3.92850
	evaluation	3.59914

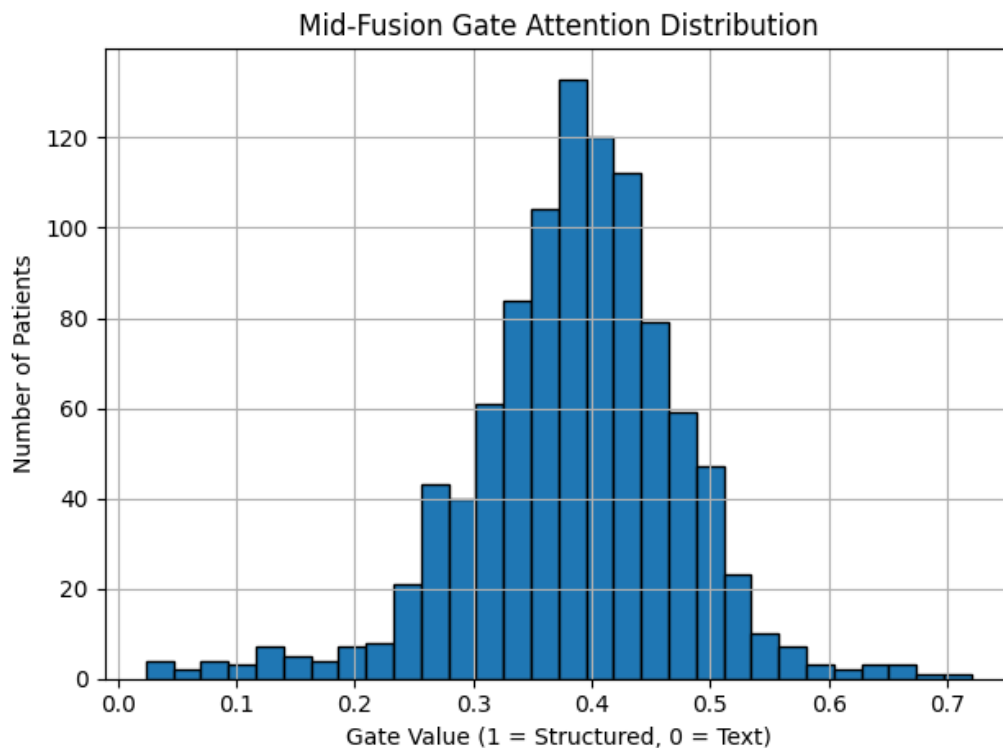


Figure 34. Mid-Fusion Gate Attention Distribution.

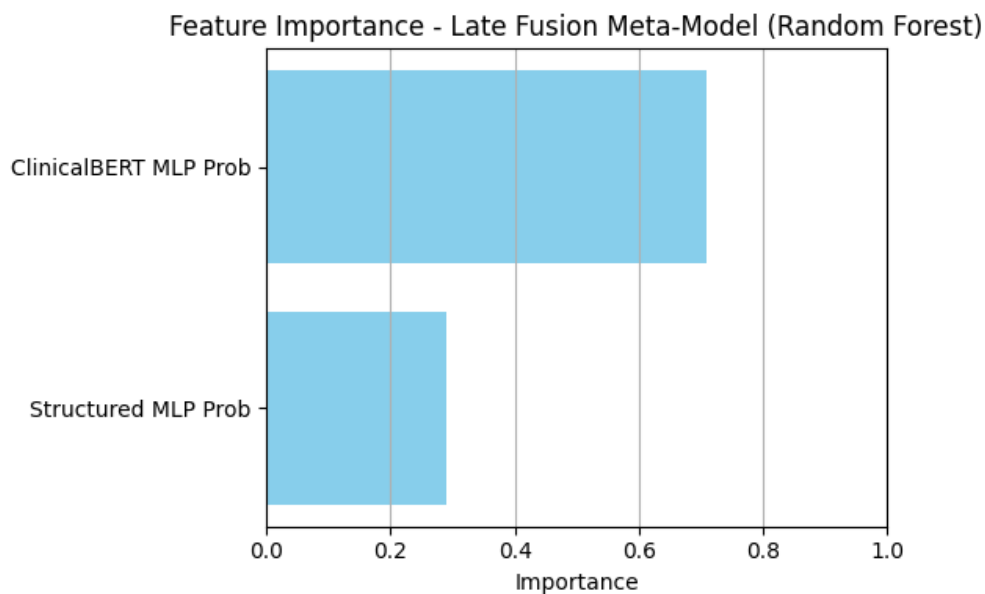


Figure 35. Late Fusion Feature Importance (Structured MLP vs ClinicalBERT MLP Probabilities).

4.6 Final Comparison and Analysis

This section presents a comparative overview of all developed models, spanning structured-only baselines, deep learning architectures, and multimodal fusion approaches.

Classical machine learning models (Logistic Regression, Random Forest, XGBoost) provided reasonable baselines but were limited by their reliance solely on structured features.

Deep learning models (Structured MLP, Fine-Tuned ClinicalBERT Transformer, ClinicalBERT MLP) outperformed traditional baselines, highlighting the value of learning complex patterns from both structured and unstructured data separately.

Multimodal fusion approaches (Mid-Fusion and Late-Fusion) achieved the strongest overall performance, with Late-Fusion attaining the highest AUROC (0.7608) and F1 Score (0.4774).

Importantly, models that incorporated clinical narratives (i.e., discharge summaries) consistently outperformed structured-only models, reinforcing the complementary nature of unstructured text in capturing nuanced patient factors.

While deep learning and fusion models achieved modest absolute gains in AUROC, their ability to significantly improve recall which is critical for identifying at-risk elderly patients, highlights their practical clinical value.

Finally, the ensemble-level robustness of the Late-Fusion model suggests that decision-level fusion is a sustainable and interpretable strategy, particularly for deployment in heterogeneous healthcare environments. Full comparison of all results is displayed below in Table 7.

Table 9. Final Model Comparison Table (Structured MLP, ClinicalBERT MLP on Fine-Tuned Embeddings, ClinicalBERT Fine-Tuned, Late Fusion RF, Hybrid Fusion MLP)

Model	Threshold	Accuracy	Precision	Recall	F1 Score	AUROC	PR AUC
Logistic Regression	0.4968	0.5513	0.2536	0.6405	0.3634	0.5973	0.2572
Random Forest	0.1720	0.6297	0.2958	0.6174	0.4000	0.6655	0.3448
XGBoost	0.0395	0.6001	0.2840	0.6579	0.3968	0.6677	0.3507
Structured MLP	0.4866	0.6185	0.2898	0.6263	0.3963	0.6656	0.3407
ClinicalBERT MLP	0.4942	0.6966	0.3557	0.6384	0.4568	0.7418	0.4350
ClinicalBERT Fine-Tuned	0.4891	0.6589	0.3149	0.5121	0.3899	0.6539	0.3487
Late Fusion (RF)	0.5710	0.7449	0.4042	0.5829	0.4774	0.7608	0.4661
Hybrid Fusion (MLP)	0.4055	0.7059	0.3656	0.6409	0.4656	0.7517	0.4457

4.7 Clinical Relevance

Accurately predicting 30-day hospital readmissions among elderly patients carries significant clinical, operational, and societal implications. The findings of this study have several important takeaways for real-world clinical practice:

Interpretability for Clinical Trust

Interpretability was prioritised throughout model development. Baseline feature importance analysis, SHAP explanations, Integrated Gradients visualization, and gating analysis of fusion models ensured that key predictors were identifiable and understandable to clinicians. Models that emphasise post-discharge healthcare utilization, medication burden, and clinical narrative signals align well with known risk factors in geriatric care, fostering potential clinical trust in predictive outputs.

Sustainable Impact Through Fusion Models

Fusion architectures demonstrated that combining structured EHR fields with clinical narratives provides more complete patient risk profiles without necessitating disruptive changes to hospital IT systems. By leveraging routinely collected discharge summaries and structured data, these models offer a sustainable path to improvement. Additionally, reducing preventable readmissions through earlier risk identification could generate significant cost savings for healthcare systems, particularly given the high burden of elderly readmissions on hospital resources.

Generalizability and Practicality

The use of simple but effective feature engineering for structured data, combined with universal discharge summaries for text inputs, enhances the generalisability of the developed models. This design minimises reliance on institution-specific variables and overly complex preprocessing pipelines, improving the feasibility of deployment across diverse healthcare settings. In particular, the Late-Fusion approach, using a lightweight Random Forest meta-classifier, offers a practical and interpretable solution that balances flexibility with strong predictive performance.

Focused Benefits for Elderly Populations

Given the unique vulnerability of elderly patients to fragmented care and readmission, multimodal models that dynamically adapt to missing information sources (e.g., incomplete structured data) are especially valuable. By drawing from both structured and unstructured signals, the models improve robustness and sensitivity in identifying high-risk individuals within this complex population.

4.8 Conclusion of Findings and Results

In summary, this chapter presented a comprehensive evaluation of all developed models, from classical machine learning baselines to advanced deep learning and multimodal fusion architectures. Consistent evaluation metrics and interpretability analyses demonstrated the progression of model performance and clinical relevance across different approaches. The findings highlight the substantial benefits of incorporating both structured and unstructured data for elderly readmission prediction. These results set the foundation for the next chapter, which will critically discuss the implications, limitations, and future directions of this work.

Chapter 5 - Discussion

5.1 Summary of Key Findings

Classical machine learning models provided reasonable baselines but were limited in capturing the complexity of elderly patient trajectories. Ensemble methods like Random Forest and XGBoost improved performance through modelling non-linear feature interactions, but structured data alone constrained predictive power.

Deep learning models, including a structured MLP and fine-tuned ClinicalBERT, consistently outperformed traditional baselines by extracting richer feature representations. The lightweight MLP trained on ClinicalBERT embeddings offered a computationally efficient alternative with minimal performance trade-off.

Fusion architectures achieved the highest predictive performance. Mid-fusion models leveraged dynamic attention gating to balance structured and unstructured signals, while late-fusion models used a Random Forest meta-classifier for decision-level combination. Late-fusion models achieved the best AUROC (0.7608) and F1 Score (0.4774).

Explainability analyses, using SHAP, Integrated Gradients, and attention gating, confirmed that both structured predictors (e.g., post-discharge healthcare utilization, medication burden) and narrative signals contributed meaningfully to predictions. Fusion models offered not just improved accuracy but also enhanced robustness and clinical interpretability.

In sum, sustainable, interpretable, and patient-centred predictive models are achievable through thoughtful integration of structured and unstructured EHR data, with strong potential for reducing readmissions among elderly patients.

5.2 Discussion of Key Results

This study demonstrated the predictive advantage of integrating structured EHR data with clinical narratives for 30-day elderly readmissions.

Classical Models

Logistic Regression struggled with non-linear patterns despite regularization and feature interaction terms. Random Forest and XGBoost captured higher-order interactions but remained limited by the information content of structured fields alone.

Deep Learning Models

Structured MLP models captured complex feature relationships more effectively, improving recall, which is critical for clinical deployment. Fine-tuned ClinicalBERT further demonstrated the value of unstructured discharge summaries by capturing subtle clinical and social signals. This aligns with Huang et al. (2019), who demonstrated the value of BERT-based models for clinical notes. This study extends that work through attention pooling and chunked inputs tailored for long, section-filtered discharge summaries. The lightweight MLP on ClinicalBERT embeddings offered a practical option for resource-constrained settings. To better capture the semantic structure of lengthy clinical narratives, attention pooling was incorporated into the ClinicalBERT architecture in place of standard [CLS] pooling. This mechanism allowed the model to dynamically emphasise contextually important tokens across long discharge summaries, improving its ability to extract clinically relevant representations. Prior research, such as Li et al. (2021), has demonstrated that attention-based pooling mechanisms can outperform static pooling strategies by better modelling token importance in clinical texts. In this study, attention pooling led to consistent gains in performance, particularly within the mid-fusion and ClinicalBERT-only models, reinforcing the importance of fine-grained token-level weighting in clinical NLP tasks. This finding is aligned with recent efforts to enhance transformer-based models in healthcare by introducing adaptive token aggregation strategies tailored to noisy, long-form EHR narratives.

Fusion Architectures

Fusion models achieved the highest performance. Mid-fusion dynamically balanced information from structured and narrative inputs per patient, while late-fusion meta-models robustly combined predictions. The late-fusion Random Forest demonstrated strong AUROC and practical interpretability, well-suited for clinical workflows.

Clinical Priorities

Recall was prioritised over precision across models to minimise the risk of missing high-risk elderly patients. Interpretability techniques consistently highlighted clinically relevant predictors, supporting trust and auditability.

Comparison to Prior Literature

The late-fusion model's AUROC of 0.7608 compares favourably to prior elderly readmission studies (typical AUROCs 0.65-0.70), confirming that multimodal integration can advance current predictive capabilities.

5.3 Implications for Clinical Practice

The findings of this study have several implications for elderly care.

Interpretability for Clinical Trust

Explainability was prioritised through SHAP, Integrated Gradients, and gating analyses, ensuring predictive outputs are transparent and clinically aligned. Structured predictors like post-discharge healthcare utilization and medication burden matched known risk factors, enhancing clinician trust.

Sustainable Impact via Fusion Models

Fusion architectures demonstrated that combining structured EHR fields with routine discharge summaries enables predictive improvements without major IT overhauls. This approach supports sustainable deployment and cost savings, especially given the financial burden of elderly readmissions.

Generalizability and Practicality

The models' reliance on standard structured data and routine narratives, rather than institution-specific variables, improves generalisability. The late-fusion Random Forest is particularly suited for practical deployment across diverse healthcare settings.

Patient-Centred Benefits

By flexibly drawing on structured or narrative data depending on availability, multimodal models enhance robustness and sensitivity, which is critical for addressing elderly patients' fragmented healthcare histories. In sum, these models offer practical, sustainable tools for reducing readmissions and improving elderly care continuity.

5.4 Limitations of the Study

While this study demonstrated promising results in predicting 30-day hospital readmissions among elderly patients, several limitations must be acknowledged.

First, although MIMIC-IV provides a diverse patient population, it remains a single-centre academic dataset. External validation on broader, multi-institutional datasets is necessary to confirm generalisability across healthcare systems, demographics, and documentation practices. The quality and structure of clinical notes, particularly discharge summaries, can vary between clinicians and institutions. While section filtering and attention pooling techniques improved robustness, variability in narrative quality remains a potential source of noise that could affect model performance in real-world deployment. This study focused exclusively on 30-day readmission as the outcome variable. Although this is a widely accepted clinical metric, it does not distinguish between avoidable and unavoidable readmissions, nor does it capture longer-term patient outcomes. Future studies could explore more granular outcome labels to better contextualise predictions. Despite the use of SMOTEENN resampling and threshold tuning to mitigate class imbalance, there remains a possibility of hidden biases. Patients with richer documentation (i.e., more complete, or detailed discharge notes) may have been implicitly favoured by the models, raising concerns about equitable performance across different patient subgroups. The computational demands of transformer fine-tuning, particularly ClinicalBERT, pose challenges for real-time deployment. Although lightweight alternatives like Late-Fusion models ease this burden, resource limitations may still hinder adoption in some institutions. While this study prioritised interpretability, true clinical trust would require prospective validation and human-in-the-loop evaluation, involving direct feedback from practicing clinicians regarding the usefulness and understandability of model outputs.

In summary, while the models presented are promising, careful external validation, continuous monitoring for biases, and practical deployment considerations are essential to ensure safe and equitable clinical translation.

5.5 Future Work

Building upon the promising results and acknowledging the study's limitations, several avenues for future research are proposed to strengthen the clinical applicability and scientific contribution of predictive models for elderly hospital readmissions.

First, external validation across multiple hospitals is critical. Applying the developed models to national datasets or diverse healthcare systems would provide more rigorous evidence of generalisability. Such validation should include both academic and community hospitals to capture a wider spectrum of patient demographics, clinical practices, and documentation standards.

Second, exploration of lighter transformer architectures represents a practical opportunity. Models such as DistilClinicalBERT or Longformer could offer comparable predictive performance while significantly reducing computational demands. Fine-tuning these models, particularly using resource-efficient strategies like LoRA (Low-Rank Adaptation), would improve feasibility for hospitals with limited infrastructure.

Third, future studies could investigate advanced multimodal fusion techniques. While mid and late-fusion approaches demonstrated impressive performance, deeper integration methods, such as cross-modal attention networks or co-attentional transformers, may better model the complex interplay between structured and unstructured clinical information, particularly for elderly populations with fragmented data.

Fourth, there is potential to expand interpretability techniques beyond traditional SHAP and Integrated Gradients. Emerging methods like Concept Activation Vectors (CAVs) or counterfactual explanations could bridge the gap between abstract model outputs and clinically meaningful concepts, offering clinicians even clearer insights into model decision-making.

Fifth, future work should prioritise clinical integration studies. Pilot deployment of these models in real-world hospital workflows would allow for iterative refinement, user feedback collection, and the measurement of downstream impacts, such as reduction in readmission rates, improved discharge planning, or healthcare cost savings.

Finally, patient-centred outcome measures could complement the current focus on 30-day readmissions. Incorporating patient-reported outcomes, post-discharge quality of life, or longitudinal follow-up data could provide a more comprehensive evaluation of the true clinical benefit of predictive modelling efforts.

In summary, future work should strive to balance further technical innovation with real-world usability, ensuring that predictive models for elderly readmission not only improve accuracy but also meaningfully support clinicians, patients, and healthcare systems.

5.6 Summary of Discussion

This study demonstrated that combining structured EHR features, and unstructured clinical narratives significantly enhances the prediction of 30-day hospital readmissions among elderly patients. Through systematic development and evaluation of classical machine learning, deep learning, and fusion models, the research advanced both predictive performance and clinical applicability. Key contributions included the creation of interpretable fusion architectures that balanced accuracy with real-world feasibility. The Late-Fusion model achieved the highest performance (AUROC 0.7608) while maintaining clinician-friendly interpretability, leveraging both structured clinical data and rich discharge narratives. Throughout, the study emphasised interpretability, generalisability, and clinical relevance, which are critical pillars for trustworthy AI in healthcare. By addressing the complexity of elderly care trajectories and focusing on sustainable solutions, this research offers a practical contribution toward reducing hospital readmissions and improving patient outcomes. The findings lay a solid foundation for scaling, validating, and operationalizing multimodal predictive models across diverse healthcare environments.

Chapter 6 - Conclusion

At the conclusion of this study, it is important to reflect on the key findings, contributions, and broader significance of the developed models. This chapter summarises the main outcomes, highlights their clinical and operational relevance, acknowledges limitations, and outlines opportunities for future research aimed at advancing sustainable, patient-centred predictive modelling in elderly care.

6.1 Restatement of Research Purpose

This thesis aimed to enhance the prediction of 30-day hospital readmissions among elderly patients by developing advanced machine learning models that integrate both structured electronic health record (EHR) features and unstructured clinical narratives. Throughout the study, interpretability, generalisability, and clinical relevance were prioritised, ensuring that predictive performance improvements would be meaningful and actionable within real-world healthcare environments.

6.2 Major Contributions

Classical machine learning models (Logistic Regression, Random Forest, XGBoost) provided reasonable baselines but remained constrained by the limitations of structured data in representing elderly patient complexity. Fine-tuning ClinicalBERT on discharge summaries captured nuanced risk factors like frailty and social vulnerabilities, demonstrating the predictive value of narrative text beyond structured fields. Hybrid fusion models achieved the highest performance, with mid-fusion dynamically balancing structured and unstructured inputs, and late-fusion ensemble models attaining the best AUROC (0.7608) and F1 Score (0.4774). This study made interpretability a central pillar, employing SHAP, Integrated Gradients, and attention gate analyses to ensure that model decisions are aligned with clinical intuition. By focusing on transparency and clinical trust, the models developed here move beyond mere predictive accuracy to meaningful, actionable tools for elderly care.

6.3 Broader Implications

The broader implications of this research extend into multiple domains of healthcare AI and clinical practice. This study reinforces the value of multimodal approaches, as elderly patients

often have fragmented records requiring flexible integration of structured and narrative data to build equitable, generalisable models. Furthermore, the emphasis on sustainable, non-disruptive data usage, leveraging routinely collected discharge summaries and standard structured features, demonstrates a clear path to real-world implementation. Hospitals and care systems can enhance predictive capabilities without requiring costly overhauls of their IT infrastructures or additional data burden, offering a sustainable improvement pathway.

This study supports the goals of SDG 3 (Good Health and Well-Being) and SDG 13 (Climate Action) by advancing sustainable, data-driven healthcare practices. By reducing avoidable hospital readmissions through early risk detection, the proposed models not only improve patient outcomes and continuity of care for elderly populations but also minimise unnecessary resource use, such as bed occupancy, medical transport, and repeat diagnostic procedures. This aligns with broader efforts to lower the healthcare sector's environmental footprint. Moreover, the model's reliance on routinely collected EHR and discharge data ensures cost-effective implementation without disrupting existing workflows, offering a scalable solution with health and operational benefits.

From a clinical operations standpoint, predictive models that prioritise recall and interpretability can better support targeted interventions, reduce preventable readmissions, and enhance continuity of care for vulnerable elderly populations. In a healthcare landscape increasingly driven by value-based care and outcome metrics, such tools are both timely and essential. The findings contribute to the growing body of evidence that narrative data, when appropriately processed and integrated, holds substantial untapped potential for improving risk stratification and decision support across diverse areas of medicine.

6.4 Final Reflections

This thesis demonstrated that the thoughtful fusion of structured EHR data and unstructured clinical narratives, guided by a commitment to interpretability and clinical trust, can meaningfully advance the field of hospital readmission prediction for elderly patients. By carefully balancing predictive performance, real-world feasibility, generalisability, and patient-centred design, the models developed here offer a pragmatic contribution to healthcare AI. Moving forward, continued efforts toward broader validation, deeper fusion methods, and real-world clinical integration will be essential to fully realise the impact of these technologies. However, the foundation laid by this study provides a strong and credible starting point: one that emphasises that predictive models for vulnerable populations must be not only accurate but also understandable, adaptable, and capable of delivering measurable clinical and financial benefits. In doing so, this research advances the vision of AI systems that are not merely tools for prediction, but true partners in delivering more sustainable, efficient, and reliable healthcare.

Bibliography

Aldhoayan, M. D., & Khayat, A. M. (2022). Leveraging Advanced Data Analytics to Predict the Risk of All-Cause Seven-Day Emergency Readmissions. *Cureus*.

<https://doi.org/10.7759/cureus.27630>

Alsentzer, E., Murphy, J. R., Boag, W., Weng, W.-H., Jin, D., Naumann, T., & McDermott, M. B. A. (2019). *Publicly Available Clinical BERT Embeddings*. <http://arxiv.org/abs/1904.03323>

Ashfaq, A., Sant'Anna, A., Lingman, M., & Nowaczyk, S. (2019). Readmission prediction using deep learning on electronic health records. *Journal of Biomedical Informatics*, 97.

<https://doi.org/10.1016/j.jbi.2019.103256>

Baltrušaitis, T., Ahuja, C., & Morency, L.-P. (2019). Multimodal machine learning: A survey and taxonomy. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 41(2), 423–443. <https://doi.org/10.1109/TPAMI.2018.2798607>

Batista, G. E. A. P. A., Prati, R. C., & Monard, M. C. (2004). A study of the behaviour of several methods for balancing machine learning training data. *ACM SIGKDD Explorations Newsletter*, 6(1), 20–29. <https://doi.org/10.1145/1007730.1007735>

Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32.

<https://doi.org/10.1023/A:1010933404324>

Brom, H., Margo Brooks Carthon, J., Ikeaba, U., & Chittams, J. (2020). Leveraging Electronic Health Records and Machine Learning to Tailor Nursing Care for Patients at High Risk for Readmissions. *Journal of Nursing Care Quality*, 35(1), 27–33.

<https://doi.org/10.1097/NCQ.0000000000000412>

Carvalho, R. M. S., Oliveira, D., & Pesquita, C. (2023). Knowledge Graph Embeddings for ICU readmission prediction. *BMC Medical Informatics and Decision Making*, 23(1).

<https://doi.org/10.1186/s12911-022-02070-7>

Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 785–794). ACM. <https://doi.org/10.1145/2939672.2939785>

Chiu, C. C., Wu, C. M., Chien, T. N., Kao, L. J., & Li, C. (2024). Predicting ICU Readmission from Electronic Health Records via BERTopic with Long Short Term Memory Network Approach. *Journal of Clinical Medicine*, 13(18). <https://doi.org/10.3390/jcm13185503>

Cox, D. R. (1958). The Regression Analysis of Binary Sequences. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 20(2), 215–232. <https://doi.org/10.1111/j.2517-6161.1958.tb00292.x>

Deschepper, M., Eeckloo, K., Vogelaers, D., & Waegeman, W. (2019). A hospital wide predictive model for unplanned readmission using hierarchical ICD data. *Computer Methods and Programs in Biomedicine*, 173, 177–183. <https://doi.org/10.1016/j.cmpb.2019.02.007>

Eckert, C., Nieves-Robbins, N., Spieker, E., Louwers, T., Hazel, D., Marquardt, J., Solveson, K., Zahid, A., Ahmad, M., Barnhill, R., McKelvey, T. G., Marshall, R., Shry, E., & Teredesai, A. (2019). Development and Prospective Validation of a Machine Learning-Based Risk of Readmission Model in a Large Military Hospital. *Applied Clinical Informatics*, 10(2), 316–325. <https://doi.org/10.1055/s-0039-1688553>

Huang, K., Altosaar, J., & Ranganath, R. (2019). *ClinicalBERT: Modeling Clinical Notes and Predicting Hospital Readmission*. <http://arxiv.org/abs/1904.05342>

Huang, Y., Talwar, A., Chatterjee, S., & Aparasu, R. R. (2021). Application of machine learning in predicting hospital readmissions: a scoping review of the literature. *BMC Medical Research Methodology*, 21(1). <https://doi.org/10.1186/s12874-021-01284-z>

Johnson, A. E. W., Pollard, T. J., Shen, L., Li-Wei, H. L., Feng, M., Ghassemi, M., ... & Mark, R. G. (2023). MIMIC-IV (version 2.2). PhysioNet. <https://doi.org/10.13026/a3wn-hq05>

Khetan, V., Rizvi, M. I. H., Huber, J., Bartusiak, P., Sacaleanu, B., & Fano, A. (2021). *MIMICause: Representation and automatic extraction of causal relation types from clinical notes*. <http://arxiv.org/abs/2110.07090>

Li, X., Cui, M., Li, J., Bai, R., Lu, Z., & Aickelin, U. (2021). A hybrid medical text classification framework: Integrating attentive rule construction and neural network. *Neurocomputing*, 443, 345–355. <https://doi.org/10.1016/j.neucom.2021.02.069>

Lin, T. Y., Goyal, P., Girshick, R., He, K., & Dollar, P. (2017). Focal Loss for Dense Object Detection. *Proceedings of the IEEE International Conference on Computer Vision, 2017-October*, 2999–3007. <https://doi.org/10.1109/ICCV.2017.324>

Lin, Y. W., Zhou, Y., Faghri, F., Shaw, M. J., & Campbell, R. H. (2019a). Analysis and prediction of unplanned intensive care unit readmission using recurrent neural networks with long shortterm memory. *PLoS ONE*, 14(7). <https://doi.org/10.1371/journal.pone.0218942>

Lin, Y. W., Zhou, Y., Faghri, F., Shaw, M. J., & Campbell, R. H. (2019b). Analysis and prediction of unplanned intensive care unit readmission using recurrent neural networks with long shortterm memory. *PLoS ONE*, 14(7). <https://doi.org/10.1371/journal.pone.0218942>

Lo, Y. T., Liao, J. C. hen, Chen, M. H., Chang, C. M., & Li, C. te. (2021). Predictive modelling for 14-day unplanned hospital readmission risk by using machine learning algorithms. *BMC Medical Informatics and Decision Making*, 21(1). <https://doi.org/10.1186/s12911-021-01639-y>

Loreto, M., Lisboa, T., & Moreira, V. P. (2020). Early prediction of ICU readmissions using classification algorithms. *Computers in Biology and Medicine*, 118. <https://doi.org/10.1016/j.compbiomed.2020.103636>

Loutati, R., Ben-Yehuda, A., Rosenberg, S., & Rottenberg, Y. (2024). Multimodal Machine Learning for Prediction of 30-Day Readmission Risk in Elderly Population. *American Journal of Medicine*, 137(7), 617–628. <https://doi.org/10.1016/j.amjmed.2024.04.002>

Low, L. L., Lee, K. H., Hock Ong, M. E., Wang, S., Tan, S. Y., Thumboo, J., & Liu, N. (2015). Predicting 30-Day Readmissions: Performance of the LACE Index Compared with a Regression Model among General Medicine Patients in Singapore. *BioMed Research International*, 2015. <https://doi.org/10.1155/2015/169870>

Lundberg, S., & Lee, S.-I. (2017). *A Unified Approach to Interpreting Model Predictions*. <http://arxiv.org/abs/1705.07874>

Meng, Q., Lai, K. W., Wang, J., & Hegselmann, S. (n.d.). *Development and validation of an interpretable 3 day intensive care unit readmission prediction model using explainable boosting machines.*

Mesgarpour, M., Chausalet, T., & Chahed, S. (2017). Ensemble Risk Model of Emergency Admissions (ERMER). *International Journal of Medical Informatics*, 103, 65–77.
<https://doi.org/10.1016/j.ijmedinf.2017.04.010>

Michailidis, P., Dimitriadou, A., Papadimitriou, T., & Gogas, P. (2022). Forecasting Hospital Readmissions with Machine Learning. *Healthcare (Switzerland)*, 10(6).
<https://doi.org/10.3390/healthcare10060981>

Moerschbacher, A., & He, Z. (2023). Building Prediction Models for 30-Day Readmissions Among ICU Patients Using Both Structured and Unstructured Data in Electronic Health Records. *Proceedings - 2023 IEEE International Conference on Bioinformatics and Biomedicine, BIBM 2023*, 4368–4373. <https://doi.org/10.1109/BIBM58861.2023.10385612>

Mohanty, S. D., Lekan, D., McCoy, T. P., Jenkins, M., & Manda, P. (2022). Machine learning for predicting readmission risk among the frail: Explainable AI for healthcare. *Patterns*, 3(1).
<https://doi.org/10.1016/j.patter.2021.100395>

Nguyen, O. K., Makam, A. N., Clark, C., Zhang, S., Xie, B., Velasco, F., Amarasingham, R., & Halm, E. A. (2016). Predicting all-cause readmissions using electronic health record data from the entire hospitalization: Model development and comparison. *Journal of Hospital Medicine*, 11(7), 473–480. <https://doi.org/10.1002/jhm.2568>

Nguyen, O. K., Washington, C., Clark, C. R., Miller, M. E., Patel, V. A., Halm, E. A., & Makam, A. N. (2021). Man vs. Machine: Comparing Physician vs. Electronic Health Record–Based Model Predictions for 30-Day Hospital Readmissions. *Journal of General Internal Medicine*, 36(9), 2555–2562. <https://doi.org/10.1007/s11606-020-06355-3>

Rumelhart, D. E., Hinton, G. E., & Williams, R. J. (1986). Learning representations by back-propagating errors. *Nature*, 323(6088), 533–536. <https://doi.org/10.1038/323533a0>

Saleh, S. N., Makam, A. N., Halm, E. A., & Nguyen, O. K. (2020). Can we predict early 7-day readmissions using a standard 30-day hospital readmission risk prediction model? *BMC Medical Informatics and Decision Making*, 20(1). <https://doi.org/10.1186/s12911-020-01248-1>

Sheetrit, E., Brief, M., & Elisha, O. (2023). Predicting unplanned readmissions in the intensive care unit: a multimodality evaluation. *Scientific Reports*, 13(1). <https://doi.org/10.1038/s41598-023-42372-y>

Shi, K., Ho, V., Song, J. J., Bechler, K., & Chen, J. H. (n.d.). *Predicting Unplanned 7-day Intensive Care Unit Readmissions with Machine Learning Models for Improved Discharge Risk Assessment*.

Talwar, A., Lopez-Olivo, M. A., Huang, Y., Ying, L., & Aparasu, R. R. (2023). Performance of advanced machine learning algorithms over logistic regression in predicting hospital readmissions: A meta-analysis. *Exploratory Research in Clinical and Social Pharmacy*, 11. <https://doi.org/10.1016/j.rcsop.2023.100317>

Wang, S., & Zhu, X. (2022). Nationwide hospital admission data statistics and disease-specific 30-day readmission prediction. *Health Information Science and Systems*, 10(1). <https://doi.org/10.1007/s13755-022-00195-7>

Wong, A. H., Whitfill, T., Ohuabunwa, E. C., Ray, J. M., Dziura, J. D., Bernstein, S. L., & Taylor, R. A. (2021). Association of Race/Ethnicity and Other Demographic Characteristics with Use of Physical Restraints in the Emergency Department. *JAMA Network Open*, 4(1), E2035241. <https://doi.org/10.1001/jamanetworkopen.2020.35241>

Xiao, C., Ma, T., Dieng, A. B., Blei, D. M., & Wang, F. (2018). Readmission prediction via deep contextual embedding of clinical concepts. *PLoS ONE*, 13(4). <https://doi.org/10.1371/journal.pone.0195024>

Zhang, X., Barnes, S., Golden, B., & Smith, P. (2021). A continuous-time Markov model for estimating readmission risk for hospital inpatients. *Journal of Applied Statistics*, 48(1), 41–60. <https://doi.org/10.1080/02664763.2019.1709810>

Zhou, H., Della, P. R., Roberts, P., Goh, L., Dhaliwal, S. S., & Dhaliwal, S. (n.d.). *Utility of models to predict 28-day or 30-day unplanned hospital readmissions: an updated systematic review*. <https://doi.org/10.1136/bmjopen-2016>

Appendix

All code used in this study, including data preprocessing scripts, model training pipelines, and evaluation procedures, is publicly available at the following GitHub repository [here](#).